数学物理方法 (简明复分析)

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2025 年 4 月 24 日

摘要

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关键词：复变函数.

定义 0.1 (可微的). 一个复变函数 $f(z)$ 在定义域 R 上单值, 称其在点 z 处是可微的 *(differentiable)*[1][2] , 若下述导数 *(derivative)* 存在

$$f'(z) = \lim_{\Delta z \to 0} \left[\frac{f(z+\Delta z) - f(z)}{\Delta z} \right]$$

定义 0.2 (解析, 全纯). 如果复变函数 f 在区域 *(单连通)*$S \subseteq R$ *(R 为其定义域)* 上处处可微, 则称其在 S 上解析 *(或全纯)(analytic)*.

定理 0.3 (Cauchy-Riemann 关系). 若 f 在区域 S 上解析, 那么 $f(x+\mathrm{i}y) = u(x,y) + \mathrm{i}v(x,y)$ 满足[3][4]

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

推论 0.4. 若 $f(x+\mathrm{i}y) = u(x,y) + \mathrm{i}v(x,y)$ 在区域 S 上解析, 那么 u, v 均是 *Laplace* 方程的解, 即

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

推论 0.5. 若 $f(x+\mathrm{i}y) = u(x,y) + \mathrm{i}v(x,y)$ 在区域 S 上解析, 那么 ∇u 与 ∇v 正交且等长, 即

$$\nabla u \cdot \nabla v = 0, \quad |\nabla u| = |\nabla v|.$$

定义 0.6 (复级数). 形如

$$f(z) = \sum_{n \geqslant 0} a_n z^n = \sum_{n \geqslant 0} a_n r^n \exp(\mathrm{i}n\theta).$$

其中 z 是复变量, a_n 是复数.

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[1]在验证可微性时因该严格用定义.

[2]若验证不可微, 一般是从导数延不同方向的值不同.

[3]一种记忆方法是联想如下矩阵 $\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$.

[4]Cauchy-Riemann 关系往往用来证明函数在某区域上不解析, 用反证法.

定理 0.7 (收敛半径)**.** 收敛半径 R 有如下计算公式[5]

$$\frac{1}{R} = \lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$$

定义 0.8. 有如下常用级数

$$\exp z = \sum_{n\geqslant 0} \frac{z^n}{n!}$$

定理 0.9. $\exp\colon (\mathbb{C},+) \to (\mathbb{C},\times)$ 是群同态.

$$\exp(z_1+z_2) = \exp(z_1)\exp(z_2).$$

定义 0.10.

$$a^z = \exp(z\ln a), \quad a\in\mathbb{R}, z\in\mathbb{C}.$$

定理 0.11.

$$\exp(\mathrm{i}y) = \cos y + \mathrm{i}\sin y.$$

推论 0.12.

$$\exp z = (\exp x)(\cos y + \mathrm{i}\sin y).$$

定义 0.13.

$$\cos z = \frac{\exp(\mathrm{i}z)+\exp(-\mathrm{i}z)}{2}, \quad \sin z = \frac{\exp(\mathrm{i}z)-\exp(-\mathrm{i}z)}{2\mathrm{i}}.$$

定义 0.14. 俯角主值 $\operatorname{Arg} z = \theta \in (-\pi,\pi], \quad z = r\exp(\mathrm{i}\theta)$.

定义 0.15. 对于多值方程 $\exp w = z$ 有如下解

$$w = \operatorname{Ln} z = \operatorname{Ln}(r\exp[\mathrm{i}(\theta+2k\pi)]) = \ln r + \mathrm{i}(\theta+2k\pi), \quad k\in\mathbb{Z}.$$

[5]这里 $|\cdot|$ 是求模运算.

定义 0.16.

$$\ln z = \ln r + i\theta, \quad -\pi < \theta \leqslant \pi.$$

定义 0.17.

$$t^z = \exp(z \operatorname{Ln} t), \quad t, z \in \mathbb{C}.$$

定理 0.18. *$t \in \mathbb{C}$ 有 n 个 n 次方根.*

$$t^{1/n} = \exp\left(\frac{1}{n}\operatorname{Ln} t\right) = r^{1/n}\exp\left[\mathrm{i}\frac{(\theta + 2k\pi)}{n}\right], \quad k = 0, \cdots, n-1.$$

定义 0.19 (支点, 割线)**.** *如果当 z 沿着 z_0 的充分小领域中的任何简单闭曲线绕一圈时, 多值函数的值就从一支编导另一支, 那么称 z_0 为多值函数的一个支点. 割线将多值函数定于域割为单值的.*

定义 0.20 (奇点)**.** *f 在全平面上不解析的点称为奇点 (singularity).*

命题 0.21. *奇点[6]常常分为孤立与非孤立[7]. 支点, 可去奇点[8], 极点[9], 本性奇点[10]均是奇点.*

定义 0.22 (分式线性变换)**.** *形如 $w = T(z) = \dfrac{az+b}{cz+d}$ 的变换.[11]*

定理 0.23. *分式线性变换可以由三点 (及其对应像点) 唯一确定.*

定理 0.24.

$$w' = T'(z) = \frac{ad - bc}{(cz+d)^2}$$

[6]无穷远处的奇点往往通过替换 $\xi = 1/z \to 0$ 来分析.

[7]在该点的任意去心领域内都存在其他奇点.

[8]补充极限即可, 一般是孤立的.

[9]Laurent 展开有有限负次项.

[10]Laurent 展开做不到有限负次项.

[11]在分式线性映射中, 往往要求边界映为边界, 内部映为内部.

定义 0.25 (共形映照)**.**

$$w = g(z) = r(x, y) + \mathrm{i}s(x, y).$$

定理 0.26. *共形映照保连续, 保夹角, 放大率与方向无关, 保解析.*

定理 0.27. *若 z_0 在 $\operatorname{Im} z > 0$ 上一点, 那么共形映照*

$$w = (\exp \mathrm{i}\phi)\frac{z - z_0}{z - \bar{z}_0}.$$

将上半平面映为单位圆盘.

定义 0.28 (复积分)**.** *复积分可由下述实积分定义.*

$$\begin{aligned}\int_C f(z)\mathrm{d}z &= \int_C u\mathrm{d}x - \int_C v\mathrm{d}y + \mathrm{i}\int_C u\mathrm{d}y + \mathrm{i}\int_C v\mathrm{d}x \\ &= \int_\alpha^\beta u\frac{\mathrm{d}x}{\mathrm{d}t}\mathrm{d}t - \int_\alpha^\beta v\frac{\mathrm{d}y}{\mathrm{d}t}\mathrm{d}t + \mathrm{i}\int_\alpha^\beta u\frac{\mathrm{d}y}{\mathrm{d}t}\mathrm{d}t + \mathrm{i}\int_\alpha^\beta v\frac{\mathrm{d}x}{\mathrm{d}t}\mathrm{d}t.\end{aligned}$$

定理 0.29. *若在曲线 C 上 $|f(z)| \leqslant M$, L 为曲线 C 的长度, 那么*

$$\left|\int_C f(z)\mathrm{d}z\right| \leqslant ML.$$

定理 0.30 (Cauchy)**.** *若 $f(z)$ 是解析函数, 且 $f'(z)$ 在闭曲线 C 内及 C 上每点连续, 则*

$$\oint_C f(z)\mathrm{d}z = 0.$$

推论 0.31. *设 C 和 γ 是复数平面中的两条闭曲线, γ 充分小且完全包含在 C 内. 若函数 $f(z)$ 在两曲线间的区域内解析, 则可以更换复积分的路径.*

$$\oint_C f(z)\mathrm{d}z = \oint_\gamma f(z)\mathrm{d}z.$$

定理 0.32 (最大模原理)**.** *f 是域 D 中的非常数的全纯函数, 那么 $|f(z)|$ 不可能在 D 中取到最大值.*

定理 0.33 (Cauchy 积分公式)**.** *若 $f(z)$ 在闭曲线 C 内及 C 上解析, z_0 是 C 内一点, 则*

$$f(z_0) = \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(z)}{z - z_0} \mathrm{d}z.$$

推论 0.34.

$$f^{(n)}(z_0) = \frac{n!}{2\pi \mathrm{i}} \oint_C \frac{f(z)\mathrm{d}z}{(z - z_0)^{n+1}}$$

推论 0.35. *设 $f(z)$ 在以 z_0 为中心, 半径为 R 的圆 C 内及 C 上解析. 若在圆周上 $|f(z)| \leq M$, 其中 M 为常数, 则有关于高阶导数的模长的估计公式:*

$$|f^{(n)}(z_0)| \leqslant \frac{M \cdot n!}{R^n}.$$

定理 0.36 (Taylor 级数)**.** *设函数 $f(z)$ 在以 z_0 为中心、半径为 R 的圆 C 内及 C 上解析，则 $f(z)$ 可以在 C 内展开成泰勒级数*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{f^{(n)}(z_0)}{n!}.$$

定理 0.37 (恒等定理)**.** *若函数 $f(z)$ 和 $g(z)$ 在区域 R 内解析, 且在 R 的某个子区域 S 内满足 $f(z) = g(z)$, 则 $f(z) = g(z)$ 在整个区域 R 上成立.*

定理 0.38 (Laurent 级数)**.** *若函数 $f(z)$ 在以 z_0 为中心的两个同心圆 C_1 和 C_2 之间的环形区域 R 内解析, 且 z 为 R 内任意一点, 则 $f(z)$ 可以在环形区域 R 内展开成洛朗级数.*

$$f(z) = \sum_{n=-\infty}^{+\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi \mathrm{i}} \oint \frac{f(z)}{(z - z_0)^{n+1}} \mathrm{d}z.$$

定义 0.39 (残数)**.** *残数*[12]*指 Laurent 展开中的系数 a_{-1}.*

定理 0.40. *一般函数 f 在 m 阶极点 z_0 处的留数为:*

$$\mathrm{Res}(f, z_0) = a_{-1} = \lim_{z \to z_0} \left\{ \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}} \left[(z - z_0)^m f(z) \right] \right\} \Big/ (m-1)!$$

[12]又称留数 (residue).

定理 0.41. 如果 $f=\dfrac{g}{h}$, g 和 h 在 z_0 处全纯, 且 $g(z_0)\neq 0$, $h(z_0)=0$, $h'(z_0)\neq 0$. 则

$$\operatorname{Res}(f,z_0)=\frac{g(z_0)}{h'(z_0)}.$$

定理 0.42 (残数定理)**.** 设 D 是复平面上的一个有界区域, 它的边界 γ 由一条或若干条简单闭曲线组成. 如果 f 在 D 中除去孤立奇点 $z_1,\cdots,z_n$ 外是全纯的, 在闭域 $\bar{D}$ 上去除 $z_1,\cdots,z_n$ 外是连续的, 那么

$$\int_\gamma f(z)\mathrm{d}z=2\pi\mathrm{i}\sum_{k=1}^{n}\operatorname{Res}(f,z_k).$$

命题 0.43. 对于形如 $\displaystyle\int_0^{2\pi}F(\cos\theta,\sin\theta)\mathrm{d}\theta$, 一般做如下处理.

$$\begin{cases}\sin\theta=-\dfrac{1}{2}\mathrm{i}(z-z^{-1})\\ \cos\theta=\dfrac{1}{2}(z+z^{-1})\\ \mathrm{d}\theta=-\mathrm{i}z^{-1}\mathrm{d}z\\ C:z=\exp\mathrm{i}\theta\end{cases}$$

命题 0.44. 设 f 在上半平面 $\{z\colon \operatorname{Im}z>0\}$ 中除去 $a_1,\cdots,a_n$ 外是全纯的, 在 $\{z\colon \operatorname{Im}z\geqslant 0\}$ 中除去 $a_1,\cdots,a_n$ 外是连续的. 如果 $\displaystyle\lim_{z\to\infty}zf(z)=0$, 那么

$$\int_{-\infty}^{\infty}f(x)\mathrm{d}x=2\pi\mathrm{i}\sum_{k=1}^{n}\operatorname{Res}(f,a_k).$$

定理 0.45. 设 f 在上半平面 $\{z\colon \operatorname{Im}z>0\}$ 中除去 $a_1,\cdots,a_n$ 外是全纯的, 在 $\{z\colon \operatorname{Im}z\geqslant 0\}$ 中除去 $a_1,\cdots,a_n$ 外是连续的. 如果 $\displaystyle\lim_{z\to\infty}f(z)=0$, 那么对任意 $\alpha>0$, 有

$$\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}\alpha x}f(x)\mathrm{d}x=2\pi\mathrm{i}\sum_{k=1}^{n}\operatorname{Res}(\mathrm{e}^{\mathrm{i}\alpha z}f(z),a_k).$$

参考文献

教材.

A 作业与练习

附作业与综合练习 (测验 1 与作业 7(第八周, 练习 2)).

数学物理方法测验 1

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2025 年 4 月 1 日

摘要

24.1 单变量复变函数: 理解解析函数的概念. 验证给定函数是否为解析函数.

24.2 柯西-黎曼方程: 导出柯西-黎曼方程. 使用柯西-黎曼方程证明其它结论.

24.3 复变量的幂级数: 计算复变量的幂级数的收敛半径.

24.4 一些初等函数: 掌握指数函数, 对数函数和幂函数的定义和性质.

24.5 多值函数与分枝: 找出给定复变函数的分枝点和单值区域.

24.6 复变函数的奇点与零点: 区分可去奇点, 极点和本性奇点. 计算极点和零点的阶数.

题目 1.

a) 讨论极限 $\lim\limits_{z\to z_0}\dfrac{\operatorname{Im} z-\operatorname{Im} z_0}{z-z_0}$ 是否存在.

b) 设函数 $f(z)=y^2-x^2+ax+by+\mathrm{i}(cxy+3x+2y)$ 在复平面上处处解析, 求 a,b,c 的值.

c) 设函数 $f(z)=axy+\mathrm{i}(bx^2+y^2)$ 在复平面上处处解析, 求 a,b 的值.

解答.

1. 我们记 $z=x+\mathrm{i}y$, $z_0=x_0+\mathrm{i}y_0$. 那么原极限化为

$$\begin{aligned}\lim_{z\to z_0}\frac{y-y_0}{(x-x_0)+\mathrm{i}(y-y_0)}&=\lim_{z\to z_0}1\Big/\frac{(x-x_0)+\mathrm{i}(y-y_0)}{y-y_0}\\&=\lim_{z\to z_0}1\Big/\left(\frac{x-x_0}{y-y_0}+\mathrm{i}\right).\end{aligned}$$

我们让 z 沿 $y-y_0=m(x-x_0)\to 0$ 的方式趋于 z_0, 那么

$$\lim_{\substack{x\to x_0\\ y=y_0+m(x-x_0)\to y_0}} 1\Big/\left(\frac{x-x_0}{y-y_0}+\mathrm{i}\right)=\lim_{x\to x_0}1\Big/\left[\frac{x-x_0}{m(x-x_0)}+\mathrm{i}\right]=\frac{m}{1+\mathrm{i}m}.$$

这就表明了原极限不存在.

2. 我们记 $u(x,y)=y^2-x^2+ax+by$, $v(x,y)=cxy+3x+2y$.

   由于 f 在全复平面上解析, 那么其在全复平面上处处满足 Cauchy-Riemman 关系, 即

$$\begin{cases} u_x=v_y \quad \text{i.e.} \quad -2x+a=cx+2 \\ u_y=-v_x \quad \text{i.e.} \quad 2y+b=-cy-3 \end{cases}, \forall x,y.$$

   由 x,y 的任意性解得 $(a,b,c)=(2,-3,-2)$.

3. 我们记 $u(x,y)=axy$, $v(x,y)=bx^2+y^2$.

   由于 f 在全复平面上解析, 那么其在全复平面上处处满足 Cauchy-Riemman 关系, 即

$$\begin{cases} u_x=v_y \quad \text{i.e.} \quad ay=2y \\ u_y=-v_x \quad \text{i.e.} \quad ax=-2bx \end{cases}, \forall x,y.$$

   由 x,y 的任意性解得 $(a,b)=(2,-1)$.

题目 2.

a) 设函数 $f(z)=u(x,y)+\mathrm{i}v(x,y)$ 在区域 R 内解析, 且 $v=u^2$. 证明 $f(z)$ 在 R 内是常数.

b) 设 (r,θ) 是复数的极坐标, 设 $f(z)=u(r,\theta)+\mathrm{i}v(r,\theta)$ 解析, 证明

$$r\frac{\partial u}{\partial r}=\frac{\partial v}{\partial \theta},\quad \frac{\partial u}{\partial \theta}=-r\frac{\partial v}{\partial r}.$$

解答.

1. 我们将 $v=u^2$ 代入得 $f=u+\mathrm{i}u^2$.

由于 f 在 R 内解析, 则在 R 内处处满足 Cauchy-Riemman 关系, 即

$$\begin{cases} u_x = 2u \cdot u_y \\ u_y = -2u \cdot u_x \end{cases}, R \quad \Longrightarrow \begin{cases} u_x = -4u^2 \cdot u_x \\ u_y = -4u^2 \cdot u_y \end{cases}, R.$$

上述关系有解

$$\begin{cases} u^2 = -1/4, \quad \text{or} \quad u_x = 0. \\ u^2 = -1/4, \quad \text{or} \quad u_y = 0. \end{cases}$$

由于 u 是 $\mathbb{R}^2$ 上的函数 $\mathbb{R}^2 \xrightarrow{u} \mathbb{R}$, 那么 $\mathbb{R}^2 \xrightarrow{u^2} \mathbb{Z}_{\geqslant 0}$, 这样只有唯一解:

$$(u_x, u_y) = (0, 0)$$

由于区域是连通的 (数学分析中开域, 闭域, 或者开域连同其一部分界点所成的点集统称区域), 那么由 $(u_x, u_y) = (0, 0)$ 可以得到 u 为常值 (函数), 从而 f 在 R 内是常值. □

(R 的连通性是必要的, 否则 u 做不到在整个 R 上为常值, 只能做到在每个连通的子集上是常值, 而这些常值却做不到一定是相等的.)

2. 我们先给出坐标变换

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \Longrightarrow \left(\frac{\partial(x,y)}{\partial(r,\theta)}\right)_{2\times 2} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}.$$

从而

$$\left(\frac{\partial(r,\theta)}{\partial(x,y)}\right) = \left(\frac{\partial(x,y)}{\partial(r,\theta)}\right)^{-1} = \begin{pmatrix} \cos\theta & \sin\theta \\ -r^{-1}\sin\theta & r^{-1}\cos\theta \end{pmatrix}. \tag{1}$$

由于 $z = x + \mathrm{i}y$, f 是解析的, 那么 f 在所有解析的 (r, θ) 处都满足 Cauchy-Riemman 关系:

$$\begin{cases} u_x = v_y \\ u_y = -v_x \end{cases} \quad \text{i. e.} \begin{cases} \dfrac{\partial u}{\partial r}\dfrac{\partial r}{\partial x} + \dfrac{\partial u}{\partial \theta}\dfrac{\partial \theta}{\partial x} = \dfrac{\partial v}{\partial r}\dfrac{\partial r}{\partial y} + \dfrac{\partial v}{\partial \theta}\dfrac{\partial \theta}{\partial y} \\ \dfrac{\partial u}{\partial r}\dfrac{\partial r}{\partial y} + \dfrac{\partial u}{\partial \theta}\dfrac{\partial \theta}{\partial y} = -\dfrac{\partial v}{\partial r}\dfrac{\partial r}{\partial x} - \dfrac{\partial v}{\partial \theta}\dfrac{\partial \theta}{\partial x} \end{cases}, \forall r, \theta.$$

将(1)代入上式并整理得到

$$\begin{cases} \left(\dfrac{\partial u}{\partial r} - \dfrac{1}{r}\dfrac{\partial v}{\partial \theta}\right)\cos\theta = \left(\dfrac{\partial v}{\partial r} + \dfrac{1}{r}\dfrac{\partial u}{\partial \theta}\right)\sin\theta \\ \left(\dfrac{\partial u}{\partial r} - \dfrac{1}{r}\dfrac{\partial v}{\partial \theta}\right)\sin\theta = \left(\dfrac{\partial v}{\partial r} + \dfrac{1}{r}\dfrac{\partial u}{\partial \theta}\right)(-\cos\theta) \end{cases}, \forall r, \theta.$$

由于解析点 (r,θ) 的任意性, 上式只有一个结果

$$r\frac{\partial u}{\partial r}=\frac{\partial v}{\partial \theta},\quad \frac{\partial u}{\partial \theta}=-r\frac{\partial v}{\partial r}.$$

□

题目 3. 初等函数的计算

a) $\sqrt[4]{-16}$.

b) $\sqrt[\mathrm{i}]{2\mathrm{i}}$.

c) $\mathrm{Ln}(3-3\mathrm{i})$.

d) 解方程 $\cos(z)=3$.

e) 用 $x+\mathrm{i}y$, $r(\cos\theta+\mathrm{i}\sin\theta)$, $r\mathrm{e}^{\mathrm{i}\theta}$ 三种方式表示复数 $\mathrm{e}^{1+\mathrm{i}}$. 在复平面上画出这个复数

解答.

1.

$$\begin{aligned}\sqrt[4]{-16}=&(-16)^{1/4}\exp\left[\frac{1}{4}\mathrm{Ln}(-16)\right]=\exp\left\{\frac{1}{4}\mathrm{Ln}\left[16\exp\left(\mathrm{i}\pi+2k\mathrm{i}\pi\right)\right]\right\}\\=&\exp\left[\frac{1}{4}\left(\ln 16+\mathrm{i}\pi+2k\mathrm{i}\pi\right)\right]=\exp\left(\ln 2+\mathrm{i}\frac{\pi}{4}+\mathrm{i}\frac{k\pi}{2}\right)\\=&2\cdot\left[\cos\left(\frac{\pi}{4}+\frac{k\pi}{2}\right)+\mathrm{i}\sin\left(\frac{\pi}{4}+\frac{k\pi}{2}\right)\right],\quad k\in\mathbb{Z}.\end{aligned}$$

故 $\sqrt[4]{-16}$ 有四个值, 分别是

$$\sqrt{2}\pm\mathrm{i}\sqrt{2},\quad -\sqrt{2}\pm\mathrm{i}\sqrt{2}.$$

2.

$$\begin{aligned}\sqrt[\mathrm{i}]{2\mathrm{i}}=&(2\mathrm{i})^{1/\mathrm{i}}=\exp\left[\frac{1}{\mathrm{i}}\mathrm{Ln}(2\mathrm{i})\right]=\exp\left\{-\mathrm{i}\,\mathrm{Ln}\left[2\exp\left(\mathrm{i}\frac{\pi}{2}+2k\mathrm{i}\pi\right)\right]\right\}\\=&\exp\left\{-\mathrm{i}\left[\ln 2+\mathrm{i}\left(\frac{\pi}{2}+2k\pi\right)\right]\right\}=\exp\left(\frac{\pi}{2}+2k\pi-\mathrm{i}\ln 2\right)\\=&\mathrm{e}^{\pi/2}\cdot\mathrm{e}^{2k\pi}\cdot\left[\cos\ln 2-\mathrm{i}\sin\ln 2\right],\quad k\in\mathbb{Z}.\end{aligned}$$

3.

$$\mathrm{Ln}(3-3\mathrm{i})=\mathrm{Ln}\left[3\sqrt{2}\exp\left(-\mathrm{i}\frac{\pi}{4}+2k\mathrm{i}\pi\right)\right]=\frac{1}{2}\ln 18+\mathrm{i}\left(-\frac{\pi}{4}+2k\pi\right),\quad k\in\mathbb{Z}.$$

4. 由 cos 的定义我们有

$$\cos z = \frac{\exp(\mathrm{i}z) + \exp(-\mathrm{i}z)}{2} = 3 \Longrightarrow \exp(\mathrm{i}z) + \exp(-\mathrm{i}z) = 6.$$

我们做变量代换 $\omega = \exp(\mathrm{i}z)$, 则方程化为

$$\omega + 1/\omega = 6 \Longrightarrow \omega^2 - 6\omega + 1 = 0. \quad (\omega = 0\text{显然不是解})$$

解得 $\exp(\mathrm{i}z) = \omega = 3 \pm 2\sqrt{2} \in \mathbb{R}$. 现在我们利用 ω 反解 z:

$$\mathrm{i}z = \mathrm{Ln}(\omega) = \ln\omega + 2k\mathrm{i}\pi, \quad k \in \mathbb{Z}.$$

从而

$$z = -\mathrm{i}\ln\omega + 2k\pi = 2k\pi - \mathrm{i}\ln(3 \pm 2\sqrt{2}) = 2k\pi \pm 2\mathrm{i}\ln(\sqrt{2}+1), \quad k \in \mathbb{Z}.$$

5. 这题是显然的 (图1)

$$\mathrm{e}^{1+\mathrm{i}} = \mathrm{e} \cdot \mathrm{e}^{\mathrm{i}\cdot 1} = \mathrm{e}(\cos 1 + \mathrm{i}\sin 1) = \mathrm{e}\cos 1 + \mathrm{i}(\mathrm{e}\sin 1).$$

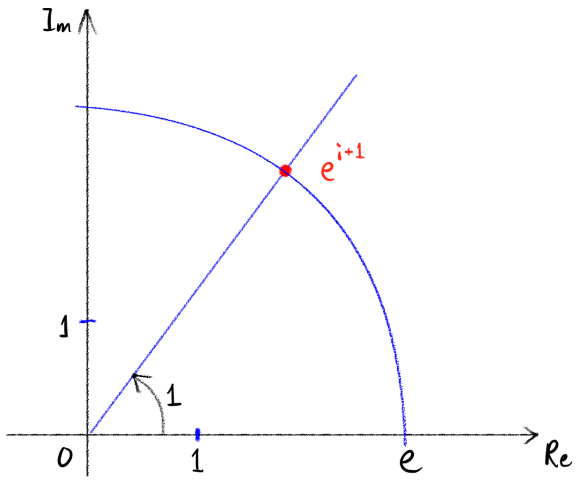

图 1: $\mathrm{e}^{\mathrm{i}+1}$ 的图像

题目 4. 求下列幂级数的收敛半径

a) $\displaystyle\sum_{n=1}^{\infty} \frac{z^n}{n^n}$.　　b) $\displaystyle\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} z^n$.　　c) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{(2n+1)!!} z^{2n}$.

解答.

1.
$$\lim_{n\to+\infty}\sqrt[n]{\left|\frac{1}{n^n}\right|}=\lim_{n\to+\infty}\frac{1}{n}=0\Longrightarrow R=+\infty.$$

2.
$$\lim_{n\to+\infty}\sqrt[n]{\left|\left(1+\frac{1}{n}\right)^{n^2}\right|}=\lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^{n}=\mathrm{e}\Longrightarrow R=\frac{1}{\mathrm{e}}.$$

3.
$$\lim_{n\to+\infty}\left|\frac{(n+1)!}{(2n+3)!!}\Big/\frac{n!}{(2n+1)!!}\right|=\lim_{n\to+\infty}\left|\frac{n}{2n+3}\right|=\frac{1}{2}\Longrightarrow R=2.$$

题目 5.

a) 找出 $f(z)=\sqrt{z(z-1)(z-2)}$ 的分支点, 并确定一条可能的分支割线.

b) 找出 $f(z)=\sqrt[3]{z(z-1)}$ 的分支点，并确定一条可能的分支割线。设某个单值解析分支的 $f(2)$ 为正实数，求 $f(-1)$ 的值。

解答.

1. 分支点是显然的: $z=0,z=1,z=2$. 这里给出一条可能的分支割线:

$$\text{实轴上 } [0,1]\cup[2,+\infty).$$

实质上有很多, 例如实轴上 $[0,+\infty)$, 实轴上 $(-\infty,0]\cup[1,2]$ 等.

2. 分支点是显然的: $z=0,z=1$. 这里给出一条可能的分支割线:

$$\text{折射线: 实轴上 } [0,1]\cup\tau.$$
$$\tau \text{ 为以 } 1\in\mathbb{R} \text{ 为起点, 方向与负虚轴一至的射线.}$$

我们先计算 $f(2)$

$$f(2)=\sqrt[3]{2}=\sqrt[3]{2}\cdot\mathrm{e}^{\mathrm{i}2k\pi/3},\quad k\in\mathbb{Z}.$$

由于 $f(2)\in\mathbb{R}_{>0}$, 那么 $f(2)=\sqrt[3]{2}\in\mathbb{R}$. 考察 $f(-1)=\sqrt[3]{2}$, 有 $|f(-1)|=|f(2)|$.

在上述分支割线的情况下, 其从上方绕过,$f(-1)$ 的辐角相比于 $f(2)$ 增加了 $2\pi/3$, 则

$$f(-1)=f(2)\mathrm{e}^{2\pi\mathrm{i}/3}=\sqrt[3]{2}\cdot\left(-\frac{1}{2}+\mathrm{i}\frac{\sqrt{3}}{2}\right)=\frac{\sqrt[3]{2}}{2}+\mathrm{i}\frac{\sqrt[3]{2}\sqrt{3}}{2}.$$

在确定分支割线后,$f(-1)$ 的值完全被确定. 不同分支割线可能导致不同的 $f(-1)$, 例如当 τ 为以 $1 \in \mathbb{R}$ 为起点, 方向与正虚轴一至的射线时, $f(-1) = \sqrt[3]{2} \cdot \left(-\frac{1}{2} - \mathrm{i}\frac{\sqrt{3}}{2}\right)$. 选择分支割线的时候应尽量不要让要研究的点 $-1, 2$ 落在分支割线上.

题目 6. 确定函数的奇点及其类型, 对无穷远点也要讨论. 若是极点, 求出阶数.

a) $\dfrac{1-\cos(z)}{z^2}$.　　b) $\dfrac{z-1}{z(z^2+4)^2}$.　　c) $\dfrac{1-\exp(z^2)}{z^4\sin(z)}$.

d) $\dfrac{1}{(z^2+i)^3}$.　　e) $\dfrac{1}{\sin(z)+\cos(z)}$.

解答. 以下皆有 $\xi := 1/z$.

1. 0 是一个奇点, 且是可去奇点. 这是由于
$$\lim_{z\to 0}\frac{1-\cos(z)}{z^2} = \frac{1}{2}$$
对于无穷远处
$$\frac{1-\cos(z)}{z^2} = \frac{1}{z^2}\left(1-1+\frac{z^2}{2!}-\frac{z^4}{4!}+\cdots\right) = \frac{1}{2}-\frac{z^2}{4!}+\cdots$$
这显然是一个本性奇点.

2. 显然 0 是一个一阶极点, $\pm 2\mathrm{i}$ 均为二阶极点.

对于无穷远显然是一个奇点.
$$\frac{z-1}{z(z^2+4)^2} = \frac{1-1/z}{(z^2+4)^2} = \frac{1-\xi}{(1/\xi^2+4)^2} \longrightarrow 0. \quad (\xi \to 0)$$
这说明无穷远是一个可去奇点.

3. 显然 $z=0$ 是一个三阶极点, $z = k\pi$ $(k \in \mathbb{Z}\setminus\{0\})$ 是一阶极点.

对于无穷远处, 研究 $\xi = x + \mathrm{i}y$ 从不同方向趋向与 0
$$\frac{1-\exp(z^2)}{z^4\sin(z)} = \xi^4\frac{1-\exp(1/\xi^2)}{\sin(1/\xi)}$$
$$\xi^4\frac{1-\exp(1/\xi^2)}{\sin(1/\xi)} \overset{y=0}{=} x^4\frac{1-\exp(1/x^2)}{\sin(1/x)} \longrightarrow \text{震荡的}. \quad (x \to 0)$$
$$\xi^4\frac{1-\exp(1/\xi^2)}{\sin(1/\xi)} \overset{x=0}{=} y^4\frac{\exp(-1/y^2)-1}{\sin(\mathrm{i}/y)} = y^4\frac{\exp(-1/y^2)-1}{\mathrm{i}\sinh(1/y)}$$
$$= -\mathrm{i}\left[\frac{y^4(1-\mathrm{e}^{y^2})}{\sinh(1/y)\mathrm{e}^{y^2}}\right] \longrightarrow 0. \quad (y \to 0)$$
这表明无穷远是一个本性奇点.(实质上还是非孤立的.)

4. 显然 $\pm(1-\mathrm{i})/\sqrt{2}$ 是三阶极点.

   对于无穷远
$$\frac{1}{(z^2+i)^3}=\frac{\xi^6}{(1+\mathrm{i}\xi^2)^3}=\longrightarrow 0. \quad (\xi\to 0)$$
   这表明无穷远是一个可去点.

5. 利用三角恒等式 $\sin(z)+\cos(z)=\sqrt{2}\sin(z+\pi/4)$ 处理分母
$$\frac{1}{\sin(z)+\cos(z)}=\frac{1}{\sqrt{2}\sin(z+\pi/4)}$$
   那么显然 $z=-\pi/4+k\pi,\ k\in\mathbb{Z}$ 是一阶极点.

   无穷远处做如下处理
$$\frac{1}{\sqrt{2}\sin(z+\pi/4)}=\frac{1}{\sqrt{2}\sin(1/\xi+\pi/4)}.$$
   观察 $\xi\to 0$ 便足以说明其是非孤立奇点.

题目 6 的注记. Page839, Line7:

'Tending to infinity' is convenient to have a unique meaning and this is provided by the following **definition**: the behaviour of $f(z)$ at **infinity** is given by that of $f(1/\xi)$ at $\xi=0$, where $\xi=1/z$.

数学物理方法作业 1

唐嘉琪 (231330131)

2025 年 3 月 11 日

Problem 1. Show that the function $f(z) = x^3 - 3xy^2 + \mathrm{i}(3x^2y - y^3)$ is differentiable for all values of z.

Solution. Taking $\Delta z = \Delta x + \mathrm{i}\Delta y$, we have

$$\begin{aligned}
&\frac{f(z+\Delta z) - f(z)}{\Delta z} \\
=&\frac{(x+\Delta x)^3 - 3(x+\Delta x)(y+\Delta y)^2 + \mathrm{i}\left(3(x+\Delta x)^2(y+\Delta y) - (y+\Delta y)^3\right)}{\Delta x + \mathrm{i}\Delta y} \\
&+ \frac{-x^3 + 3xy^2 - \mathrm{i}(3x^2y - y^3)}{\Delta x + \mathrm{i}\Delta y} \\
=&\frac{3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3 - 6xy\Delta y - 3x(\Delta y)^2 - 3\Delta x y^2 - 6\Delta x y \Delta y}{\Delta x + \mathrm{i}\Delta y} \\
&+ \frac{-3\Delta x(\Delta y)^2 + \mathrm{i}(3x^2\Delta y + 6xy\Delta x + 6x(\Delta x)\Delta y + 3y(\Delta x)^2 + 3(\Delta x)^2\Delta y)}{\Delta x + \mathrm{i}\Delta y} \\
&+ \frac{\mathrm{i}(-3y^2\Delta y - 3y(\Delta y)^2 - (\Delta y)^3)}{\Delta x + \mathrm{i}\Delta y}
\end{aligned}$$

When $\Delta z = \Delta x + \mathrm{i}\Delta y \to 0$, that is, $(\Delta x, \Delta y) \to (0,0)$. Noticed the higher - order infinitesimal terms, the above formula can be simplified as:

$$\frac{3x^2\Delta x - 6xy\Delta y - 3\Delta x y^2 + \mathrm{i}(3x^2\Delta y + 6xy\Delta x - 3y^2\Delta y)}{\Delta x + \mathrm{i}\Delta y} + o(1)$$

After further simplification, we have

$$\begin{aligned}\frac{f(z+\Delta z)-f(z)}{\Delta z} &= \frac{(3x^2+6xy\mathrm{i}-3y^2)(\Delta x+\mathrm{i}\Delta y)}{\Delta x+\mathrm{i}\Delta y}+o(1)\\ &= 3x^2+6xy\mathrm{i}-3y^2+o(1)\to 3x^2+6xy\mathrm{i}-3y^2 \quad (\Delta z\to 0)\end{aligned}$$

Since z was arbitrary, $f(z)$ is differentiable for all values of z. □

The Note of Problem 1. Noticed $f(z)=(x+\mathrm{i}y)^3=z^3$, which is a simpler method, we have

$$\begin{aligned}\frac{f(z+\Delta z)-f(z)}{\Delta z} &= \frac{(z+\Delta z)^3-z^3}{\Delta z}\\ &= \frac{3z^2\Delta z+3z(\Delta z)^2+(\Delta z)^3}{\Delta z}\to 3z^2 \quad (\Delta z\to 0)\end{aligned}$$

Hence, another why to find the unique limitation.

Problem 2. Show that the function $f(z)=x^2+\mathrm{i}y^2$ is not differentiable anywhere in the complex plane.

Solution.

$$\begin{aligned}\frac{f(z+\Delta z)-f(z)}{\Delta z} &= \frac{(x+\Delta x)^2+\mathrm{i}(y+\Delta y)^2-x^2-\mathrm{i}y^2}{\Delta x+\mathrm{i}\Delta y}\\ &= \frac{(\Delta x)^2+2x\Delta x+\mathrm{i}((\Delta y)^2+2y\Delta y)}{\Delta x+\mathrm{i}\Delta y}\\ &= \frac{2x\Delta x+\mathrm{i}2y\Delta y}{\Delta x+\mathrm{i}\Delta y}+o(1) \quad (\Delta z\to 0)\end{aligned}$$

In this case the limit will clearly depend on the direction from which $\Delta z\to 0$. Suppose $\Delta z\to 0$ along a line through z of slope m, so that $\Delta y=m\Delta x$, then

$$\lim_{\Delta z\to 0}\left[\frac{f(z+\Delta z)-f(z)}{\Delta z}\right]=\lim_{(\Delta x,\Delta y)\to(0,0)}\left[\frac{2x\Delta x+\mathrm{i}2y\Delta y}{\Delta x+\mathrm{i}\Delta y}\right]=\frac{2x+\mathrm{i}2ym}{1+\mathrm{i}m}$$

This limit is dependent on m and hence on the direction from which $\Delta z \to 0$. Since this conclusion is independent of the value of z, and hence true for all z, $f(z) = x^2 + \mathrm{i}y^2$ is nowhere differentiable. $\square$

Problem 3. Show that the function $f(z) = 1/z^2$ is analytic everywhere except at $z = 0$.

Solution.

$$f'(z) = \lim_{\Delta z \to 0} \left[\frac{1/(z+\Delta z)^2 - 1/z^2}{\Delta z} \right] = \lim_{\Delta z \to 0} \left[\frac{1}{\Delta z} \left(\frac{1}{(z+\Delta z)^2} - \frac{1}{z^2} \right) \right]$$

$$= \lim_{\Delta z \to 0} \left[-\frac{2z + \Delta z}{(z+\Delta z)^2 z^2} \right] = -\frac{2}{z^3}$$

Independently of the way in which $\Delta z \to 0$, provided $z \neq 0$. Hence $f(z)$ is analytic everywhere except at the singularity $z = 0$. $\square$

Problem 4. Let $f(z) = u(x,y) + \mathrm{i}v(x,y)$ be differentiable at the point z. Prove the Cauchy–Riemann relations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Solution. If we let $\Delta z = \Delta x + \mathrm{i}\Delta y$ then we have

$$f(z + \Delta z) = u(x + \Delta x, y + \Delta y) + \mathrm{i}v(x + \Delta x, y + \Delta y).$$

Since $f(z)$ is differentiable at the point z, the following limit exists and is unique, and is independent of the way in which $\Delta z \to 0$.

$$\lim_{(\Delta x, \Delta y) \to (0,0)} \left[\frac{u(x+\Delta x, y+\Delta y) + \mathrm{i}v(x+\Delta x, y+\Delta y) - u(x,y) - \mathrm{i}v(x,y)}{\Delta x + \mathrm{i}\Delta y} \right]$$

First, let Δz approach 0 along the direction of $\Delta y = 0$

$$\lim_{\Delta x \to 0} \left[\frac{u(x+\Delta x, y) - u(x,y)}{\Delta x} + \mathrm{i} \frac{v(x+\Delta x, y) - v(x,y)}{\Delta x} \right] = \frac{\partial u}{\partial x} + \mathrm{i} \frac{\partial v}{\partial x} \quad (1)$$

Then, let Δz approach 0 along the direction of $\Delta x = 0$

$$\lim_{\Delta y \to 0} \left[\frac{u(x, y+\Delta y) - u(x,y)}{\mathrm{i}\Delta y} + \mathrm{i} \frac{v(x, y+\Delta y) - v(x,y)}{\mathrm{i}\Delta y} \right] = \frac{1}{\mathrm{i}} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \quad (2)$$

Since the limit is unique, it can be seen from equations (1) and (2) that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

□

Problem 5. Let $f(z) = z^2$. Verify that both the real and imaginary parts of $f(z)$ are solutions of Laplace's equation in two dimensions.

Solution.

$$f(z) = z^2 = (x + \mathrm{i}y)^2 = (x^2 - y^2 + 2\mathrm{i}xy)$$

The real part and the imaginary part are $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$ respectively.

Substitute them into the Laplace equation in two dimensions respectively.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x} 2x + \frac{\partial}{\partial y}(-2y) = 2 + (-2) = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial}{\partial x} 2y + \frac{\partial}{\partial y} 2x = 0 + 0 = 0$$

□

Problem 6. Let $f(z) = z^2 = u + \mathrm{i}v$. Verify that the families of level curves $u(x, y) = c$ and $v(x, y) = c$ intersect at right angles.

Solution.

$$f(z) = z^2 = (x + \mathrm{i}y)^2 = (x^2 - y^2 + 2\mathrm{i}xy) = u + \mathrm{i}v$$

We got $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

The vectors normal to the curve $u(x, y) = c$ and $v(x, y) = c$ are respectively given by

$$\nabla u = \frac{\partial u}{\partial x}\boldsymbol{i} + \frac{\partial u}{\partial y}\boldsymbol{j} \quad \text{and} \quad \nabla v = \frac{\partial v}{\partial x}\boldsymbol{i} + \frac{\partial v}{\partial y}\boldsymbol{j}.$$

Hence, $\nabla u = 2x\boldsymbol{i} - 2y\boldsymbol{j}$ and $\nabla v = 2y\boldsymbol{i} + 2x\boldsymbol{j}$.

The equation shows that two curves are orthogonal is given by

$$\nabla u \cdot \nabla v = 2x \cdot 2y + (-2y) \cdot 2x = 0$$

$\square$

数学物理方法作业 2

唐嘉琪 (231330131)

2025 年 3 月 22 日

Problem 1. Find the values of the following expressions.

a) $\dfrac{5}{-3+4\mathrm{i}}$. b) $\left(\dfrac{2+\mathrm{i}}{3-2\mathrm{i}}\right)^2$. c) $\sqrt{\mathrm{i}}$. d) $\sqrt{1+\mathrm{i}}$.

e) $\sin \mathrm{i}$. f) $\cos \mathrm{i}$. g) $\tan(1+\mathrm{i})$. h) 2^{i}.

i) i^{i}. j) $(-1)^{2\mathrm{i}}$.

Solution.

a)

$$\frac{5}{-3+4\mathrm{i}} = \frac{5(-3-4\mathrm{i})}{(-3+4\mathrm{i})(-3-4\mathrm{i})} = \frac{-15-20\mathrm{i}}{9+16} = -\frac{3}{5} - \frac{4}{5}\mathrm{i}.$$

b)

$$\left(\frac{2+\mathrm{i}}{3-2\mathrm{i}}\right)^2 = \left[\frac{(2+\mathrm{i})(3+2\mathrm{i})}{(3-2\mathrm{i})(3+2\mathrm{i})}\right]^2 = \left(\frac{6+7\mathrm{i}+2\mathrm{i}^2}{9+4}\right)^2 = \left(\frac{4+7\mathrm{i}}{13}\right)^2$$

$$= \frac{16+56\mathrm{i}+49\mathrm{i}^2}{169} = \frac{-33+56\mathrm{i}}{169} = -\frac{33}{169} + \frac{56}{169}\mathrm{i}.$$

c)

$$\begin{aligned}\sqrt{\mathrm{i}} &= \exp\left(\frac{1}{2}\operatorname{Ln}\mathrm{i}\right) = \exp\left[\frac{1}{2}\operatorname{Ln}\left(\exp\left(\mathrm{i}\left(\frac{\pi}{2}+2k\pi\right)\right)\right)\right]\\ &= \exp\left(\frac{1}{2}\ln 1 + \mathrm{i}\frac{\pi/2+2k\pi}{2}\right) = 1^{1/2}\exp\left[\mathrm{i}\left(\frac{\pi}{4}+k\pi\right)\right]\\ &= \pm\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}\right).\end{aligned}$$

d)

$$\begin{aligned}\sqrt{1+\mathrm{i}} &= \exp\left(\frac{1}{2}\operatorname{Ln}(1+i)\right) = \exp\left[\frac{1}{2}\operatorname{Ln}\left(\sqrt{2}\exp\left(\mathrm{i}\left(\frac{\pi}{4}+2k\pi\right)\right)\right)\right]\\ &= \exp\left(\frac{1}{2}\ln\sqrt{2} + \mathrm{i}\frac{\pi/4+2k\pi}{2}\right) = \sqrt{2}^{1/2}\exp\left[\mathrm{i}\left(\frac{\pi}{8}+k\pi\right)\right]\\ &= \sqrt[4]{2}\left(\cos\frac{\pi+k\pi}{8} + \mathrm{i}\sin\frac{\pi+k\pi}{8}\right) = \pm\sqrt[4]{2}\left(\cos\frac{\pi}{8}+\sin\frac{\pi}{8}\mathrm{i}\right)\\ &= \pm\left(\frac{\sqrt{2+2\sqrt{2}}}{2}+\frac{\sqrt{2\sqrt{2}-2}}{2}\mathrm{i}\right).\end{aligned}$$

e)

$$\sin\mathrm{i} = \frac{\exp(-1)-\exp(1)}{2\mathrm{i}} = \frac{\mathrm{e}^2-1}{2\mathrm{e}}\mathrm{i}.$$

f)

$$\cos \mathrm{i} = \frac{\exp(-1) + \exp(1)}{2} = \frac{\mathrm{e}^2 + 1}{2\mathrm{e}}.$$

g)

$$\begin{aligned}
\tan(1+\mathrm{i}) =& \frac{\sin(1+\mathrm{i})}{\cos(1+\mathrm{i})} = \frac{1}{\mathrm{i}} \cdot \frac{\exp(-1+\mathrm{i}) - \exp(1-\mathrm{i})}{\exp(-1+\mathrm{i}) + \exp(1-\mathrm{i})} \\
=& \frac{1}{\mathrm{i}} \cdot \frac{\exp(-1)\exp(\mathrm{i}) - \exp(1)\exp(-\mathrm{i})}{\exp(-1)\exp(\mathrm{i}) + \exp(1)\exp(-\mathrm{i})} \\
=& \frac{1}{\mathrm{i}} \cdot \frac{(\cos 1 + \mathrm{i}\sin 1) - [\mathrm{e}^2\cos(-1) + \mathrm{i}\sin(-1)]}{(\cos 1 + \mathrm{i}\sin 1) + [\mathrm{e}^2\cos(-1) + \mathrm{i}\sin(-1)]} \\
=& -\mathrm{i} \cdot \frac{(1-\mathrm{e}^2)\cos 1 + 2\mathrm{i}\sin 1}{(1+\mathrm{e}^2)\cos 1} = \frac{2\sin 1}{(1+\mathrm{e}^2)\cos 1} + \mathrm{i}\frac{\mathrm{e}^2 - 1}{\mathrm{e}^2 + 1}.
\end{aligned}$$

h)

$$\begin{aligned}
2^{\mathrm{i}} =& \exp(\mathrm{i}\,\mathrm{Ln}\,2) = \exp[\mathrm{i}(\ln 2 + 2\mathrm{i}k\pi)] = \exp(-2k\pi + \mathrm{i}\ln 2) \\
=& \exp(-2k\pi)\exp(\mathrm{i}\ln 2) = \mathrm{e}^{2n\pi}\left(\cos\ln 2 + \mathrm{i}\sin\ln 2\right),\ n \in \mathbb{Z}.
\end{aligned}$$

i)

$$\begin{aligned}
\mathrm{i}^{\mathrm{i}} =& \exp\left(\mathrm{i}\,\mathrm{Ln}\,\mathrm{i}\right) = \exp\left[\mathrm{i}\left(\ln 1 + \mathrm{i}\left(\frac{\pi}{2} + 2k\pi\right)\right)\right] = \exp\left(-\frac{\pi}{2} - 2k\pi + \mathrm{i}\ln 1\right) \\
=& \exp\left(-\frac{2}{\pi} - 2k\pi\right)\exp(\mathrm{i}\ln 1) = \mathrm{e}^{\pi/2}\mathrm{e}^{2n\pi},\ n \in \mathbb{Z}.
\end{aligned}$$

j)

$$(-1)^{2\mathrm{i}} = \exp\left[2\mathrm{i}\,\mathrm{Ln}(-1)\right] = \exp\left[2\mathrm{i}\left(\ln 1 + \mathrm{i}\left(\pi + 2k\pi\right)\right)\right]$$

$$= \exp(-2\pi - 4k\pi)\exp(2\mathrm{i}\ln 1) = \mathrm{e}^{2\pi}\mathrm{e}^{4n\pi},\ \ n \in \mathbb{Z}.$$

Problem 2. Find the real and imaginary parts of the following complex functions.

a) $\exp(x + \mathrm{i}y)$. b) $\cos(x + \mathrm{i}y)$. c) $\sin(x + \mathrm{i}y)$. d) $\mathrm{Ln}(x + \mathrm{i}y)$.

Solution. With x and y real, we have immediately

a)

$$\exp(x + \mathrm{i}y) = \exp x \exp(\mathrm{i}y) = \mathrm{e}^x(\cos y + \mathrm{i}\sin y).$$

Hence, the real part is $\mathrm{e}^x \cos y$ and the imaginary part is $\mathrm{e}^x \sin y$.

b)

$$\begin{aligned}\cos(x + \mathrm{i}y) =& \frac{\exp(-y + \mathrm{i}x) + \exp(y - \mathrm{i}x)}{2}\\ =& \frac{\mathrm{e}^{-y}(\cos x + \mathrm{i}\sin x) + \mathrm{e}^{y}(\cos x - \mathrm{i}\sin x)}{2}\\ =& \frac{(1 + \mathrm{e}^{2y})\cos x}{2\mathrm{e}^y} + \mathrm{i}\frac{(1 - \mathrm{e}^{2y})\sin x}{2\mathrm{e}^y}.\end{aligned}$$

Hence, the real part and the imaginary part are $\dfrac{(1 + \mathrm{e}^{2y})\cos x}{2\mathrm{e}^y}$ and $\dfrac{(1 - \mathrm{e}^{2y})\sin x}{2\mathrm{e}^y}$ respectively.

c)

$$\begin{aligned}\sin(x+\mathrm{i}y) =& \frac{\exp(-y+\mathrm{i}x)-\exp(y-\mathrm{i}x)}{2\mathrm{i}}\\ =& \frac{\mathrm{e}^{-y}(\cos x+\mathrm{i}\sin x)-\mathrm{e}^{y}(\cos x-\mathrm{i}\sin x)}{2\mathrm{i}}\\ =& \frac{(1+\mathrm{e}^{2y})\sin x}{2\mathrm{e}^{y}}+\mathrm{i}\frac{(\mathrm{e}^{2y}-1)\cos x}{2\mathrm{e}^{y}}.\end{aligned}$$

Hence, the real part and the imaginary part are $\dfrac{(1+\mathrm{e}^{2y})\sin x}{2\mathrm{e}^{y}}$ and $\dfrac{(\mathrm{e}^{2y}-1)\cos x}{2\mathrm{e}^{y}}$ respectively.

d)

$$\begin{aligned}\mathrm{Ln}(x+\mathrm{i}y) =& \mathrm{Ln}\left[\sqrt{x^2+y^2}\exp\left(\mathrm{i}\left(\lim_{\xi\to x}\arctan\frac{y}{\xi}+2k\pi\right)\right)\right]\\ =& \frac{1}{2}\ln(x^2+y^2)+\mathrm{i}\left(\lim_{\xi\to x}\arctan\frac{y}{\xi}+2k\pi\right),\ k\in\mathbb{Z}\end{aligned}$$

Hence, the real part and the imaginary part are $0.5\ln(x^2+y^2)$ and $\lim\limits_{\xi\to x}\arctan\dfrac{y}{\xi}+2k\pi$ respectively.

Problem 3. Find the radii of convergence of the following Taylor series:

a) $\displaystyle\sum_{n=2}^{\infty}\frac{z^n}{\ln n}$.

b) $\displaystyle\sum_{n=1}^{\infty}\frac{n!z^n}{n^n}$.

c) $\displaystyle\sum_{n=1}^{\infty}z^n n^{\ln n}$.

d) $\displaystyle\sum_{n=1}^{\infty}\left(\frac{n+p}{n}\right)^{n^2}z^n$.

Solution.

a)

$$\lim_{n\to\infty}\left|\frac{\ln n}{\ln(n+1)}\right| = \lim_{n\to\infty}\frac{n+1}{n} = 1$$

Hence, the radio of convergence is $R = 1$.

b)

$$\lim_{n\to\infty}\left|\frac{(n+1)!\,n^n}{(n+1)^{n+1}\,n!}\right| = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n = \exp\left(\lim_{n\to\infty}\frac{-n}{n+1}\right) = \frac{1}{\mathrm{e}}$$

Hence, the radio of convergence is $R = e$.

c)

$$\lim_{n\to\infty}\sqrt[n]{|n^{\ln n}|} = \lim_{n\to\infty} n^{\frac{\ln n}{n}} = \exp\left(\lim_{n\to\infty}\frac{\ln^2 n}{n}\right) = \exp\left(\lim_{n\to\infty}\frac{2\ln n}{n}\right) = 1$$

Hence, the radio of convergence is $R = 1$.

d) Suppose $p = x + \mathrm{i}y$ be a given number.

$$\lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{n+p}{n}\right)^{n^2}\right|} = \lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{n+x}{n}+\mathrm{i}\frac{y}{n}\right)^{n^2}\right|}$$

$$= \lim_{n\to\infty}\sqrt[n]{\left[\sqrt{\left(\frac{n+x}{n}\right)^2+\left(\frac{y}{n}\right)^2}\right]^{n^2}} = \lim_{n\to\infty}\left[\left(\frac{n+x}{n}\right)^2+\left(\frac{y}{n}\right)^2\right]^{n/2}$$

$$= \lim_{n\to\infty}\exp\left(\frac{n}{2}\cdot\frac{2nx+x^2+y^2}{n^2}\right) = \begin{cases}\mathrm{e}^x, & x\neq 0\\ 1, & x = 0\end{cases}$$

Therefore, when p is a pure imaginary number, the radius of convergence is $R = 1$; when p is not a pure imaginary number, the radius of convergence is $R = 1/e^x$.

The Note of Problem 3. It was given the condition for Question d) that

"with p real" in Chinese version. And the answer becomes to $1/\mathrm{e}^p$, which ignores the cases of "p could be a pure imaginary number" with this extract condition.

Problem 4. Find an analytic function of $z = x + \mathrm{i}y$ whose imaginary part is $(y\cos y + x\sin y)\exp x$.

Solution. Let analytic function $f(z) = f(x+\mathrm{i}y) = u(x,y) + \mathrm{i}v(x,y)$ and $v(x,y) = (y\cos y + x\sin y)\exp x$.

Cause of the Cauchy-Riemman relations, we got PDEs:

$$\begin{cases} \dfrac{\partial u(x,y)}{\partial x} = \dfrac{\partial v(x,y)}{\partial y} = \mathrm{e}^x[-y\sin y + (1+x)\cos y] \\ \dfrac{\partial u(x,y)}{\partial y} = -\dfrac{\partial v(x,y)}{\partial x} = -\mathrm{e}^x[(1+x)\sin y + y\cos y] \end{cases}.$$

Let $P(x,y) = \dfrac{\partial u(x,y)}{\partial x}$ and $Q(x,y) = \dfrac{\partial u(x,y)}{\partial y}$, then we notice

$$\frac{\partial P(x,y)}{\partial y} = e^x[(-x-2)\sin y - y\cos y] = \frac{\partial Q(x,y)}{\partial x},$$

which means $P(x,y)\mathrm{d}x + Q(x,y)\mathrm{d}y = 0$ is a total derivative equation and $u(x,y)$, precisely, is its general integral.

$$\begin{aligned} \mathrm{d}u(x,y) =& \mathrm{e}^x[-y\sin y + (1+x)\cos y]\mathrm{d}x - \mathrm{e}^x[(1+x)\sin y + y\cos y]\mathrm{d}y \\ =& (\cos y\mathrm{d}\mathrm{e}^x(1+x) + \mathrm{e}^x(1+x)\mathrm{d}\cos y) \\ &+ ((-\cos y - y\sin y)\mathrm{d}\mathrm{e}^x + \mathrm{e}^x\mathrm{d}(-\cos y - y\sin y)) \\ =& \mathrm{d}\mathrm{e}^x((1+x)\cos y - \cos y - y\sin y) \\ =& \mathrm{d}\mathrm{e}^x(x\cos y - y\sin y) \end{aligned}$$

Hence, though ODE theory, we simply got $u(x,y) = \mathrm{e}^x(x\cos y - y\sin y) +$

C. Therefore we got one analytic function of z.

$$\begin{aligned} f(z) =& \mathrm{e}^x(x\cos y - y\sin y) + \mathrm{i}\mathrm{e}^x(y\cos y + x\sin y) \\ =& (x+\mathrm{i}y)\exp x(\cos y + \mathrm{i}\sin y) = z\exp z \end{aligned}$$

Problem 5. Find a function $f(z)$, analytic in a suitable part of the Argand diagram, for which

$$\operatorname{Re} f(z) = \frac{\sin 2x}{\cosh 2y + \cos 2x}.$$

Where are the singularities of $f(z)$?

Solution. Let $f(z) = u(x,y) + \mathrm{i}v(x,y)$ and $u(x,y) = \dfrac{\sin 2x}{\cosh 2y + \cos 2x}$ is known. Through Cauchy-Riemman relations, we got PDEs:

$$\begin{cases} \dfrac{\partial v(x,y)}{\partial x} = -\dfrac{\partial u(x,y)}{\partial y} = -\dfrac{2\sin 2x \sinh 2y}{(\cosh 2y + \cos 2x)^2}, \\ \dfrac{\partial v(x,y)}{\partial y} = \dfrac{\partial u(x,y)}{\partial x} = \dfrac{2\cos 2x \cosh 2y + 2}{(\cosh 2y + \cos 2x)^2}. \end{cases}$$

Let $P(x,y) = \dfrac{\partial v(x,y)}{\partial x}$ and $Q(x,y) = \dfrac{\partial v(x,y)}{\partial y}$, then we notice

$$\frac{\partial P(x,y)}{\partial y} = \frac{-4\sin 2x\cosh 2y(\cosh 2y + \cos 2x) + 8\sin 2x \sinh^2 2y}{(\cosh 2y + \cos 2x)^3} = \frac{\partial Q(x,y)}{\partial x},$$

which means $P(x,y)\mathrm{d}x + Q(x,y)\mathrm{d}y = 0$ is a total derivative equation and $v(x,y)$, precisely, is its general integral.

$$\begin{aligned} \mathrm{d}v(x,y) =& -\frac{2\sin 2x \sinh 2y}{(\cosh 2y + \cos 2x)^2}\mathrm{d}x + \frac{2\cos 2x\cosh 2y + 2}{(\cosh 2y + \cos 2x)^2}\mathrm{d}y \\ =& \mathrm{d}\frac{-\cosh 2y}{\cosh 2y + \cos 2x}. \end{aligned}$$

Hence, though ODE theory, we simply got

$$v(x,y) = \frac{-\cosh 2y}{\cosh 2y + \cos 2x} + C.$$

Therefore we got one analytic function of z.

$$\begin{aligned} f(z) =& \frac{\sin 2x}{\cosh 2y + \cos 2x} + \mathrm{i}\frac{-\cosh 2y}{\cosh 2y + \cos 2x} \\ =& \frac{\sin 2x + \mathrm{i}\cosh 2y}{\cosh 2y + \cos 2x} \end{aligned}$$

The singular points of the function $f(z)$ occur where the denominator $\cosh 2y + \cos 2x = 0$.

Since $\cosh 2y = \dfrac{e^{2y} + e^{-2y}}{2} \geqslant 1$ and $\cos 2x \in [-1, 1]$, when $\cos 2x = -1$ and $\cosh 2y = 1$ (i.e., $y = 0, 2x = (2k+1)\pi,\ k \in \mathbb{Z}$), the denominator is 0.

So the singular points are $z = (k + 1/2)\,\pi$, where $k \in \mathbb{Z}$.

Problem 6. Explain the graph of the complex function $w = \sin(z)$.

Solution. $w = \sin(z) = \dfrac{\exp(\mathrm{i}z) - \exp(-\mathrm{i}z)}{2\mathrm{i}}$.
We suppose $z = x + \mathrm{i}y$.Then we have

$$\begin{aligned} w =& \sin(x + \mathrm{i}y) = \frac{\exp(-y + \mathrm{i}x) - \exp(y - \mathrm{i}x)}{2\mathrm{i}} \\ =& \frac{-\mathrm{i}}{2} \cdot [\mathrm{e}^{-y}(\cos x + \mathrm{i}\sin x) - \mathrm{e}^{y}(\cos x - \mathrm{i}\sin x)] \\ =& \frac{(1 + \mathrm{e}^{2y})\sin x + \mathrm{i}(\mathrm{e}^{2y} - 1)\cos x}{2\mathrm{e}^{y}} \end{aligned}$$

It shows that $w = \sin(z) = \sin(z + 2k\pi),\ \forall k \in \mathbb{Z}$. For a given constant $x = x_0$, $w = w(y) = [(1 + \mathrm{e}^{2y})\sin x_0 + \mathrm{i}(\mathrm{e}^{2y} - 1)\cos x_0]/(2\mathrm{e}^{y})$. We put it into $\mathbb{R}^2$.

$$w(x, y) \colon = \left(\frac{(1 + \mathrm{e}^{2y})\sin x}{2\mathrm{e}^{y}}, \frac{(\mathrm{e}^{2y} - 1)\cos x}{2\mathrm{e}^{y}} \right)$$

Problem 7. Find the Taylor series expansion about the origin of the function $f(z)$ defined by

$$f(z) = \sum_{n=1}^{\infty} (-1)^{n+1} \sin\left(\frac{pz}{n}\right)$$

where p is a constant. Hence verify that $f(z)$ is a convergent series for all z.

The Note of Problem 7. Cancel.

数学物理方法作业 3

唐嘉琪 (231330131)

2025 年 4 月 7 日

Problem 1. Find the branch points of $f(z) = \sqrt[3]{z}$, and find a possible branch cut.

Solution. For $f(z) = \sqrt[3]{z} = z^{1/3} = r^{1/3} \exp\left(\frac{\mathrm{i}\theta}{3}\right)$, we have branch point $z = 0$ trivially. One of the suitable branch cut is

The Positive Real Axis From 0 **To** $+\infty$ **.**

Problem 2. Find the branch points of $f(z) = \sqrt{z^2 - 4}$, and hence sketch suitable arrangements of branch cuts.

Solution. $f(z) = \sqrt{z^2 - 4} = \sqrt{z+2} \cdot \sqrt{z-2}$. Then we get the branch points $z = \pm 2$ trivially. One of a common branch cut is

A Straight Line Segment From $z = -2$ **To** $z = 2$ **.**

Problem 3. Find the singularities of the functions

a) $f(z) = \dfrac{1}{1-z} + \dfrac{1}{1+z}$.

b) $f(z) = \tan(z)$.

Solution.

a) $z = \pm 1$, where $1 - z = 0$ or $1 + z = 0$.

b) $f(z) = \tan(z) = \dfrac{\sin z}{\cos z}$, $\cos z = \dfrac{\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z}}{2}$. Let $\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z} = 0$, i.e.

$$\begin{aligned} \exp(\mathrm{i}z) = -\exp(-\mathrm{i}z) &= \exp[\mathrm{i}(\pi + 2n\pi)]\exp(-\mathrm{i}z) \\ \Rightarrow z = &\pi + 2n\pi - z \Rightarrow z = \pi/2 + n\pi, \quad n \in \mathbb{Z}. \end{aligned}$$

Hence the singularities of $\tan(z)$ are $z = \pi/2 + n\pi$, $\forall n \in \mathbb{Z}$.

Problem 4. Show that $f(z) = (\sin^2 z)/z$ has a removable singularity at $z = 0$.

Solution. $z = 0$ is a singularity trivially. Now, we begin to show that it's removable.

$$\begin{aligned} f(z) = \frac{\sin^2 z}{z} &= \frac{1}{z}\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right)\sin z \\ &= \left(1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots\right) \cdot \sin z \\ \Rightarrow f(z) \to 1 \cdot 0 = 0 &\quad (z \to 0) \end{aligned}$$

which means $z = 0$ is a removable singularity of$f(z)$. $\square$

Problem 5. Find the behaviour at infinity of the functions

a) $f(z) = a + bz^{-1}$. b) $f(z) = z^3(1 + z)$. c) $f(z) = \sin(z)$.

Solution.

a) Substitute $z = 1/\zeta$, then $f(1/\zeta) = a + b\zeta$ which is analytic at $\zeta = 0$, thus f is analytic at $z = \infty$.

b) Substitute $z = 1/\zeta$, then $f(1/\zeta) = (1 + 1/\zeta)/\zeta^3 = 1/\zeta^4 + 1/\zeta^3$, thus f has a pole of order 4 at$z = \infty$.

c) Substitute $z = \dfrac{1}{\zeta}$,

$$f(z) = \sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$$

, then

$$f\left(\frac{1}{\zeta}\right) = \sin\left(\frac{1}{\zeta}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{\zeta^{2n+1}}.$$

Thus f has an essential singularity at $z = \infty$.

Problem 6. Determine the types of singularities (if any) possessed by the following functions at $z = 0$ and $z = \infty$:

a) $(z-2)^{-1}$.

b) $(1 + z^3)/z^2$.

c) $\sinh(1/z)$.

d) e^z/z^3.

e) $z^{1/2}/(1 + z^2)^{1/2}$.

Solution.

a) Only have one singularity at $z = 2$.

b) $\dfrac{1+z^2}{z^2} = \dfrac{1}{z^2} + 1 = f(z)$. Then $z = 0$ is a pole of order 2.
Substitute $z = 1/\zeta$, $f(1/\zeta) = 1/\zeta + \zeta^2$, thus $z = \infty$ is a simple pole (pole of order 1).

c) $\sinh(1/z) = \exp(1/z) - \exp(-1/z)$.

Substitute $z = 1/\zeta$, then $\sinh(\zeta) = \exp(\zeta) - \exp(-\zeta)$. We know $\zeta = \infty$ (i.e. $z = 0$) is an essential singularity of $\exp(\zeta)$, thus $z = 0$ is the essential singularity of $\sinh(1/z)$.

What's more, $\sinh(1/z)$is analytic at $\zeta = 0$ i.e. $z = \infty$ trivially from equation: $\sinh(\zeta) = \exp(\zeta) - \exp(-\zeta)$.

d) $z = 0$ is a pole of order 3 trivially.

Substitute$z = 1/\zeta$, then $e^z/z^3 = \zeta^3 \cdot e^{1/\zeta}$. Cause of $\zeta = 0$ (i.e $z = \infty$) is an essential singularity of $e^{1/\zeta} = e^t$.

Hence $z = \infty$ is an essential singularity of e^z/z^3.

e) $z = 0$ is a branch point.

Substitute $z = 1/\zeta$, $\dfrac{\sqrt{z}}{\sqrt{1+z^2}} = \dfrac{\zeta}{\sqrt{\zeta}\sqrt{\zeta^2+1}} = \dfrac{\sqrt{\zeta}}{\sqrt{1+\zeta^2}}$.

Hence $\zeta = 0$ (i.e. $z = \infty$) is a branch point as well.

Problem 7. Identify the zeros, poles and essential singularities of the following functions:

a) $\tan z$.

b) $[(z-2)/z^2]\sin[1/(1-z)]$.

c) $\exp(1/z)$.

d) $\tan(1/z)$.

e) $z^{2/3}$.

Solution. omitted

数学物理方法作业

唐嘉琪 (231330131)

2025 年 4 月 8 日

Problem 1. Prove the following properties of conformal transformations.

a) The angle between two intersecting curves in the z-plane equals the angle between the corresponding curves in the w-plane.

b) The magnification, as between the z-plane and the w-plane, of a small line element in the neighbourhood of any particular point is independent of the direction of the element.

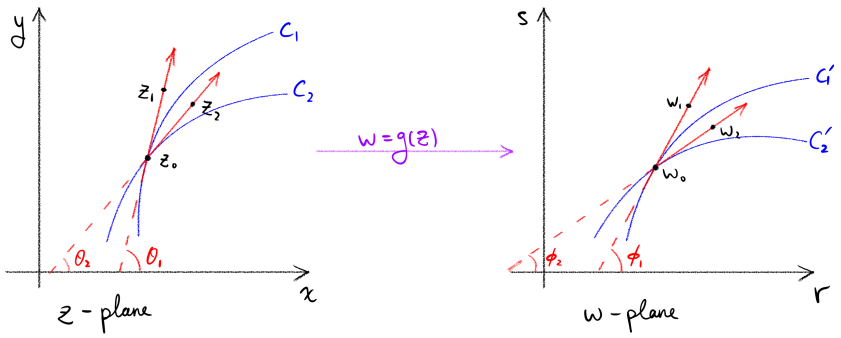

图 1: conformal transformation $w = g(z)$

Solution. We first give the figure 1 about two curves C_1 and C_2 in the z-plane, which are mapped onto C_1' and C_2' in the w-plane. Let two curves

C_1 and C_2 pass through the point z_0 in the z-plane and let z_1 and z_2 be two points on their respective tangents at z_0, each a distance ρ from z_0. The same prescription with w replacing z describes the transformed situation. We should remark that the transformed tangents may not be straight lines and the distances of w_1 and w_2 from w_0 have not yet been shown to be equal. So we suppose z_1 and z_2 are given by

$$\begin{cases} z_1 - z_0 = \rho \exp(\mathrm{i}\theta_1) \\ z_2 - z_0 = \rho \exp(\mathrm{i}\theta_2) \end{cases}$$

The corresponding descriptions in the w -plane are

$$\begin{cases} w_1 - w_0 = \rho_1 \exp(\mathrm{i}(\phi_1 + \delta\phi_1)) \\ w_2 - w_0 = \rho_2 \exp(\mathrm{i}(\phi_2 + \delta\phi_2)) \end{cases}$$

where $\delta\phi_i \to 0$ as $\rho \to 0, \quad i = 1, 2$. Since $w = g(z)$ is analytic, we have:

$$\lim_{z_1 \to z_0} \frac{w_1 - w_0}{z_1 - z_0} = \lim_{z_2 \to z_0} \frac{w_2 - w_0}{z_2 - z_0} = \left.\frac{\mathrm{d}g}{\mathrm{d}z}\right|_{z=z_0}$$

i.e

$$\lim_{\rho \to 0} \left\{ \frac{\rho_1}{\rho} \exp[\mathrm{i}(\phi_1 + \delta\phi_1 - \theta_1)] \right\} = \lim_{\rho \to 0} \left\{ \frac{\rho_2}{\rho} \exp[\mathrm{i}(\phi_2 + \delta\phi_2 - \theta_2)] \right\} = g'(z_0)$$

Hence

$$\begin{cases} \dfrac{\rho_1}{\rho} = \dfrac{\rho_2}{\rho} = |g'(z_0)| \\ \phi_1 - \theta_1 = \phi_2 - \theta_2 = \arg g'(z_0) \end{cases}$$

The first equation $\rho_1/\rho = \rho_2/\rho = |g'(z_0)|$, $\rho \to 0$ proved a). The second equation $\Leftrightarrow \phi_1 - \phi_2 = \theta_1 - \theta_2$, $\rho \to 0$ proved b). $\square$

Problem 2. Find a conformal transformation that maps the upper half of the z-plane into the interior of the unit circle in the w-plane, and maps the point $z = 1 + \mathrm{i}$ onto $w = 0$, and the point $z = 0$ onto $w = 1$.

Solution. Suppose

$$w(z;\phi,z_0) = [\exp(\mathrm{i}\phi)]\frac{z-z_0}{z-\overline{z_0}} \quad \text{s. t.} \begin{cases} w(1+\mathrm{i};\phi,z_0) = 0 \\ w(0;\phi,z_0) = 1 \end{cases}, -\pi < \phi \leqslant \pi$$

$$\text{i. e.} \begin{cases} [\exp(\mathrm{i}\phi)]\dfrac{1+\mathrm{i}-z_0}{1+\mathrm{i}-\overline{z_0}} = 0 \\ [\exp(\mathrm{i}\phi)]\dfrac{-z_0}{-\overline{z_0}} = 1 \end{cases} \Longrightarrow \begin{cases} z_0 = 1+\mathrm{i} \\ \phi = -\dfrac{\pi}{2} \end{cases}$$

i.e. $w(z) = -\mathrm{i}\dfrac{z-(1+\mathrm{i})}{z-(1-\mathrm{i})}$

Problem 3. The analytic function $w = f(z) = (1+2\mathrm{i})z + (2+\mathrm{i})$ defines a conformal transformation between complex planes. Plot the images of the real and imaginary axes from the z-plane in the w-plane.

Solution.

$$\begin{cases} z = x + \mathrm{i}y \\ w = r + \mathrm{i}s \end{cases} \Longrightarrow \begin{cases} f(x) = (2+x) + \mathrm{i}(2x+1) \\ f(\mathrm{i}y) = (2-2y) + \mathrm{i}(y+1) \end{cases}$$

then real axis mapped into line: $s = 2r - 3$; the imaginary axes mapped into line: $r + 2s - 4 = 0$.(Figure)

Problem 4. Compute the complex integral

$$\int_C z\mathrm{d}z$$

where C is the straight line segment from $z = 0$ to $z = 1 + \mathrm{i}$ in the complex plane.

Solution.

$$\int_C z\mathrm{d}z = \frac{1}{2}z^2\Big|_{z=0}^{1+\mathrm{i}} = \frac{2\mathrm{i}}{2} = \mathrm{i}.$$

Problem 5. Compute the complex integral

$$\int_C \frac{\mathrm{d}z}{z^2}$$

where C is the unit circle $|z| = 1$ traversed counterclockwise.

令 $z = \mathrm{e}^{\mathrm{i}\theta}$,

$$\int_C \frac{1}{z^2}\mathrm{d}z = \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}\theta}\mathrm{i}d\theta = -\left.\mathrm{e}^{\mathrm{i}\theta}\right|_0^{2\pi} = 0.$$

Problem 6. Compute the complex integral

$$\int_C \frac{\mathrm{d}z}{z}$$

where C is the circle $|z - 1 - \mathrm{i}| = 1$ traversed counterclockwise.

Solution.

$$z = 1 + \mathrm{i} + \cos t + \mathrm{i}\sin t = (1 + \cos t) + \mathrm{i}(1 + \sin t), \quad 0 \leqslant t \leqslant 2\pi.$$

$$
\begin{aligned}
\frac{1}{z} =& \frac{1}{(1+\cos t)+\mathrm{i}(1+\sin t)} = \frac{(1+\cos t)-\mathrm{i}(1+\sin t)}{(1+\cos t)^2+(1+\sin t)^2} \\
=& \frac{1+\cos t}{3+2\sin t+2\cos t} - \mathrm{i}\frac{1+\sin t}{3+2\sin t+2\cos t}, \quad 0 \le t \le 2\pi.
\end{aligned}
$$

$$
\begin{aligned}
\int_C \frac{\mathrm{d}z}{z} =& \int_0^{2\pi} \frac{1+\cos t}{3+2\sin t+2\cos t}(-\sin t)\mathrm{d}t + \int_0^{2\pi} \frac{1+\sin t}{3+2\sin t+2\cos t}\cos t\mathrm{d}t \\
&+ \mathrm{i}\int_0^{2\pi} \frac{1+\cos t}{3+2\sin t+2\cos t}\cos t\mathrm{d}t + \mathrm{i}\int_0^{2\pi} \frac{1+\sin t}{3+2\sin t+2\cos t}\sin t\mathrm{d}t \\
=& \int_0^{2\pi} \frac{\cos t-\sin t}{3+2\sin t+2\cos t}\mathrm{d}t + \mathrm{i}\int_0^{2\pi} \frac{1+\cos t+\sin t}{3+2\sin t+2\cos t}\mathrm{d}t
\end{aligned} \tag{1}
$$

Now, calculate the real and imaginary part respectively.

$$
\begin{aligned}
\int_0^{2\pi} \frac{\cos t-\sin t}{3+2\sin t+2\cos t}\mathrm{d}t =& \left(\int_0^{\pi} + \int_{\pi}^{2\pi}\right) \frac{\cos t-\sin t}{3+2\sin t+2\cos t}\mathrm{d}t \\
=& \int_0^{\pi} \frac{\cos t-\sin t}{3+(2\sin t+2\cos t)}\mathrm{d}t \\
&+ \int_0^{\pi} \frac{-\cos t+\sin t}{3-(2\sin t+2\cos t)}\mathrm{d}t \\
=& 4\int_0^{\pi} \frac{\sin^2 t-\cos^2 t}{5-8\sin t\cos t}\mathrm{d}t = 0
\end{aligned} \tag{2}
$$

$$
\begin{aligned}
\int_0^{2\pi} \frac{1+\cos t+\sin t}{3+2\sin t+2\cos t}\mathrm{d}t =& \int_0^{2\pi} \left(\frac{1}{2} + \frac{1}{2}\frac{1}{3+2\cos t+2\sin t}\right)\mathrm{d}t \\
=& \pi - \frac{1}{2}\int_{\pi/4}^{9\pi/4} \frac{1}{3+2\sqrt{2}\cos\theta}\mathrm{d}\theta \\
=& \pi - \frac{1}{2}\int_0^{2\pi} \frac{1}{3+2\sqrt{2}\cos\theta}\mathrm{d}\theta = \pi - \frac{1}{2}\cdot 2\pi = 0
\end{aligned} \tag{3}
$$

Obviously,(2), (3) $\Longrightarrow$ (1) $= 0$.

数学物理方法作业 5(第六周)

唐嘉琪 (231330131)

2025 年 4 月 14 日

Problem 1. Prove Cauchy's Theorem. If $f(z)$ is an analytic function, and $f'(z)$ is continuous at each point within and on a closed contour C, then

$$\oint_C f(z)\mathrm{d}z = 0.$$

Solution. K. F. RILEY, M. P. HOBSON and S.J. BENCE. Mathematical Methods for Physics and Engineering, 3rd. Cambridge University Press. Page 849

Notice that the continuity of $f'(z)$ is not necessary for the proof of Cauchy's Thm. □

Problem 2. Verify Cauchy's theorem with an example and a counter-example.

$$\oint_{|z|=1} z^2\mathrm{d}z = 0, \ \oint_{|z|=1} \frac{1}{z}\mathrm{d}z = 2\pi\mathrm{i}.$$

Solution. z^2 and $(z^2)'$ are both analytic at each point within an on the closed contour $|z = 1|$ for the first one.

The second integral cannot use the Cauchy's Thm for the point $z = 0$, calculate directly with $z = \mathrm{e}^{\mathrm{i}\pi}$.

Problem 3. Consider two closed contours C and γ in the Argand diagram, γ being sufficiently small that it lies completely within C. Show that if the function $f(z)$ is analytic in the region between the two contours then

$$\oint_C f(z)\mathrm{d}z = \oint_\gamma f(z)\mathrm{d}z.$$

Solution. K. F. RILEY, M. P. HOBSON and S.J. BENCE. Mathematical Methods for Physics and Engineering, 3rd. Cambridge University Press. Page 851, equation(24.43). □

Problem 4. Use Green's theorem

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x\mathrm{d}y = \oint_C (P\mathrm{d}x + Q\mathrm{d}y).$$

to compute the area of the disk $|z| \leqslant 1$.

Solution. The key is following equations:

$$\begin{aligned}
\text{Area} = \iint_D \mathrm{d}x\mathrm{d}y &= \oint_{|z|=1} -\frac{1}{2}y\mathrm{d}x + \frac{1}{2}x\mathrm{d}y \\
\mathrm{d}\theta = \mathrm{d}\arctan\frac{y}{x} &= \frac{x\mathrm{d}y - y\mathrm{d}x}{\underbrace{x^2 + y^2}_{=1}}
\end{aligned}$$

Problem 5. Prove Cauchy's integral formula. If $f(z)$ is analytic within and on a closed contour C and z_0 is a point within C then

$$f(z_0) = \frac{1}{2\pi\mathrm{i}} \int_C \frac{f(z)}{z - z_0}\mathrm{d}z.$$

Solution. K. F. RILEY, M. P. HOBSON and S.J. BENCE. Mathematical Methods for Physics and Engineering, 3rd. Cambridge University Press. Page 852, equation(24.46). □

Problem 6. Use Cauchy's integral formula to compute the integral

$$\oint_C \frac{\mathrm{e}^z}{z-2}\mathrm{d}z$$

where C is the circle $|z| = 3$ traversed counterclockwise.

Solution. Let$z_0 = 2$, $f(z) = \mathrm{e}^z$. Use the Cauchy Integral Formula.

Problem 7. Prove Cauchy's integral formula for $f'(z_0)$ given by

$$f'(z_0) = \frac{1}{2\pi \mathrm{i}} \int_C \frac{f(z)}{(z-z_0)^2}\mathrm{d}z.$$

Solution. K. F. RILEY, M. P. HOBSON and S.J. BENCE. Mathematical Methods for Physics and Engineering, 3rd. Cambridge University Press. Page 852, equation(24.47). □

Problem 8. Suppose that $f(z)$ is analytic inside and on a circle C of radius R centred on the point $z = z_0$. If $|f(z)| \leqslant M$ on the circle, where M is some constant, show that

$$|f^{(n)}(z_0)| \leqslant \frac{Mn!}{R^n}.$$

Solution. K. F. RILEY, M. P. HOBSON and S.J. BENCE. Mathematical Methods for Physics and Engineering, 3rd. Cambridge University Press. Page 853, equation(24.49). □

Problem 9. Prove Liouville's theorem. If $f(z)$ is analytic and bounded for all $z \in \mathbb{C}$ then f is a constant.

Solution.

$$f'(z_0) = \frac{1}{2\pi \mathrm{i}} \int_C \frac{f(z)}{(z-z_0)^2} \mathrm{d}z$$
$$\Longrightarrow |f'(z_0)| \leqslant \frac{M}{R} \longrightarrow 0 \quad (R \to +\infty).$$

Where $M = \sup\limits_{\mathbb{C}} |f(z)|$. □

Problem 10. Prove the fundamental theorem of algebra. For a complex polynomial $p_n(z)$ of degree n, the equation $p_n(z) = 0$ has precisely n complex roots. Hint. By applying Liouville's theorem to $f(z) = 1/p_n(z)$, prove that $p_n(z) = 0$ has at least one complex root. Factor out that root to obtain $p_{n-1}(z)$ and, by repeating the process, prove the above theorem.

Solution. Let $R > 0$ be large enough s.t. |z|>R, then

$$\left| \sum_{i=1}^{n} \frac{a_{n-i}}{z^i} \right| \leqslant \frac{|a_0|}{2} \Longrightarrow |p_n(z)| \geqslant |z|^n \frac{|a_n|}{2}$$
$$\text{i.e. } |f(z)| = \frac{1}{|p_n(z)|} \leqslant \frac{2}{|a_n| R^n}.$$

For f is continous on the compact set $|z| \leqslant R$. Its easy to get that

$$|f(z)| \leqslant \max\left\{ M, \frac{2}{|a_n| R^n} \right\}, \quad M \geqslant \sup_{|z| \leqslant R} |f(z)|. \quad (\forall z \in \mathbb{C})$$

If $p_n(z)$ has no zero points i.e. $f(z)$ has no zero points. Then $f(z)$ is analytic for all C. Through Liouville's Thm we can confirm that f is a constant. Thus $p_n(z)$ has at least one complex root, say $z_1 \in \mathbb{C}$.

Then use math. induction got the result. □

Problem 11. Use Cauchy's integral formula to compute the contour integral

$$\oint_{|z|=2} \frac{\sin(z)}{(z-1)^2(z+1)} dz$$

where the contour $|z| = 2$ is traversed counterclockwise.

Solution. Substitute $\oint_{|z|=2}$ by $\oint_{|z+1|=\varepsilon} + \oint_{|z-1|=\varepsilon}$.

Then use integral formula got the result obviously.

Problem 12. Show that, if a is a positive real constant, the function $\exp(\mathrm{i}az^2)$ is analytic and $\to 0$ as $|z| \to \infty$ for $0 < \arg z \leqslant \pi/4$. By applying Cauchy's theorem to a suitable contour prove that

$$\int_0^\infty \cos(ax^2)dx = \sqrt{\frac{\pi}{8a}}.$$

Solution. 'Composition preserves analyticity' solves the analyticity of $\exp(\mathrm{i}az^2)$. Substitute z with $r\mathrm{e}^{\mathrm{i}\theta}, (0 < \theta \leqslant \pi/4)$ got $\to 0$ as $|z| \to \infty$ for $0 < \arg z \leqslant \pi/4$.

We define

$$\begin{aligned}
\Gamma_1 &:= \left\{\exp(\mathrm{i}ax^2) \colon 0 < x \leqslant R, x \text{ form } 0 \text{ to } R\right\} \\
\Gamma_2 &:= \left\{\exp(\mathrm{i}az^2) \colon \arg z = \frac{\pi}{4}, 0 < |z| \leqslant R, |z| \text{ from } R \text{ to } 0\right\} \\
\Gamma_3 &:= \left\{\exp(\mathrm{i}az^2) \colon |z| = R, 0 \leqslant \arg z \leqslant \frac{\pi}{4}, \arg z \text{ from } 0 \text{ to } \frac{\pi}{4}\right\}
\end{aligned}$$

Then through Cauchy Thm we know that

$$\oint_{\Gamma := \sum_i^3 \Gamma_i} \exp(\mathrm{i}az^2)\mathrm{d}z = 0$$

and its obviously that

$$\left|\oint_{\Gamma_3} \exp(\mathrm{i}az^2)\mathrm{d}z\right| \leqslant \frac{\pi R}{4} \cdot \mathrm{e}^{-aR^2} \longrightarrow 0 \quad (R \to +\infty)$$

Hence

$$\left(\oint_{\Gamma_1} + \oint_{\Gamma_2}\right) \exp(\mathrm{i}az^2)\mathrm{d}z = 0 \quad (R \to +\infty)$$

i.e.

$$\int_0^{+\infty} \exp(\mathrm{i}ax^2)\mathrm{d}x = \mathrm{e}^{\mathrm{i}\frac{\pi}{4}} \int_0^{\pi} \exp(at^2)\mathrm{d}t = \mathrm{e}^{\mathrm{i}\frac{\pi}{4}} \cdot \frac{1}{2} \cdot \sqrt{\frac{\pi}{a}}$$

Comparing with the Real parts got the result.

Problem 13. The function $f(z) = \sqrt{1-z^2}$ of the complex variable z is defined to be real and positive on the real axis in the range $-1 < x < 1$. Using cuts running along the real axis for $1 < x < +\infty$ and $-\infty < x < -1$, show how $f(z)$ is made single-valued and evaluate it on the upper and lower sides of both cuts. Use these results and a suitable contour in the complex z-plane to evaluate the integral

$$I = \int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}}.$$

Confirm your answer by making the substitution $x = \sec\theta$.

Solution. Omitted

数学物理方法作业 6(第七周)

唐嘉琪 (231330131)

2025 年 4 月 22 日

Problem 1. [Taylor series] If $f(z)$ is analytic inside and on a circle C of radius R centred on the point $z = z_0$, and z is a point inside C, then

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where a_n is given by $f^{(n)}(z_0)/n!$.

Solution. Since $f(z)$ is analytic inside and on C. Through Cauchy's Formula to write $f(z)$ as

$$f(z) = \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(\xi)}{\xi - z} \mathrm{d}\xi. \tag{1}$$

Where ξ lies on C. Notice that

$$\begin{aligned}
\frac{1}{\xi - z} =& \frac{1}{(\xi - z_0) - (z - z_0)} = \frac{1}{\xi - z_0} \cdot \frac{1}{1 - (z - z_0)/(\xi - z_0))} \\
=& \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n \\
\Rightarrow (1) =& \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(\xi)}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n \mathrm{d}\xi = \frac{1}{2\pi \mathrm{i}} \sum_{n=0}^{\infty} (z - z_0)^n \oint_C \frac{f(\xi)}{(\xi - z_0)^{n+1}} \mathrm{d}\xi \\
=& \frac{1}{2\pi \mathrm{i}} \sum_{n=0}^{\infty} (z - z_0)^n \cdot \frac{2\pi \mathrm{i} f^{(n)}(z_0)}{n!} = \sum_{n=0}^{\infty} a_n (z - z_0)^n
\end{aligned}$$

Problem 2. [Identity Thm] Show that if $f(z)$ and $g(z)$ are analytic in some region R, and $f(z) = g(z)$ within some subregion S of R, then $f(z) = g(z)$ throughout R.

Solution. $h(z) := f(z) - g(z)$. Obviously h is analytic.

For all $z_0 \in S$, we have $h(z) = h(z_0) + h'(z_0)(z - z_0) + \cdots$.

For h is analytic in R, we got h is convergent inside some circle C that extends at least as far as the nearest part of the boundary of R.

Since $z_0 \in S$, we have

$$h(z_0) = h'(z_0) = \cdots = h^{(0)}(z_0) = \cdots = 0$$

Then $h(z) = 0$ for all z inside circle C.

Hence we can choose a new point inside C and repeat the process, which could get the result.

Problem 3. [Laurent seires] If $f(z)$ is analytic in a region R between two circles C_1 and C_2 centred on the point $z = z_0$, and z is a point inside R, then

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n,$$

where a_n is given by $a_n = \dfrac{1}{2\pi i} \oint \dfrac{f(z)}{(z - z_0)^{n+1}} \mathrm{d}z$.

Solution. If z_0 is an analytic point, then f can expand in a Taylor series.

If z_0 is a pole of order p, then $g(z) := (z - z_0)^p f(z)$ can expand in a Taylor series.

$$g(z) = \sum_{n=0}^{\infty} b_n (z - z_0)^n \Longrightarrow f(z) = \sum_{n=-p}^{+\infty} a_n (z - z_0)^n$$

where

$$\begin{cases} b_n = \dfrac{g^{(n)}(z_0)}{n!} = \dfrac{1}{2\pi \mathrm{i}} \oint \dfrac{g(z)}{(z-z_0)^{n+1}} \mathrm{d}z \\ a_n = \dfrac{1}{2\pi \mathrm{i}} \oint \dfrac{g(z)}{(z-z_0)^{n+1+p}} \mathrm{d}z = \dfrac{1}{2\pi \mathrm{i}} \oint \dfrac{f(z)}{(z-z_0)^{n+1}} \mathrm{d}z \end{cases}$$

If z_0 is an essential singularity, then it is not possible to find such a lowest value of $-p$, which means the principle part may contain an infinite number of terms, so that

$$f(z) = \sum_{n=-\infty}^{+\infty} a_n (z-z_0)^n, \quad a_n = \frac{1}{2\pi \mathrm{i}} \oint \frac{f(z)}{(z-z_0)^{n+1}} \mathrm{d}z.$$

Problem 4. Find the Laurent series of

$$f(z) = \frac{1}{z(z-2)^3}$$

about the singularities $z = 0$ and $z = 2$ (separately). Hence verify that $z = 0$ is a pole of order 1 and $z = 2$ is a pole of order 3, and find the residue of $f(z)$ at each pole.

Solution.

$$\begin{aligned} f(z) =& \frac{1}{-8z(1-z/2)^3} = -\frac{1}{8z} \sum_{n=0}^{+\infty} \frac{(n+2)(n+1)}{2} \left(\frac{z}{2}\right)^n \\ =& -\frac{1}{8z} \sum_{n=0}^{+\infty} \frac{n^2+3n+2}{2^{n+1}} z^n, \quad (0 < |z| < 2). \end{aligned}$$

It's obviously that the residue of $f(z)$ at $z = 0$ is $-1/8$.

$$f(z) = \frac{1}{2(z-2)^3} \cdot \frac{1}{1+(z-2)/2} = \frac{1}{2(z-2)^3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2}{2}\right), \quad (0 < |z| < 2).$$

Then the residue of $f(z)$ at $z = 1$ is $1/8$.

Problem 5. Suppose that $f(z)$ has a pole of order m at the point $z = z_0$. By considering the Laurent series of $f(z)$ about z_0, derive a general expression for the residue $R(z_0)$ of $f(z)$ at $z = z_0$. Hence evaluate the residue of the function at the point $z = i$,

$$f(z) = \frac{\exp \mathrm{i}z}{(z^2+1)^2}.$$

Solution. Page 857,858

$$\mathrm{Res}(f, z_0) = a_{-1} = \lim_{z \to z_0} \left\{ \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}} \left[(z - z_0)^m f(z) \right] \right\} \Big/ (m-1)!$$

Problem 6. [Residue Thm] Let $f(z)$ be continuous within and on a closed contour C and analytic, except for a finite number of poles, within C. Then we have the residue formula

$$\oint_C f(z)\mathrm{d}z = 2\pi\mathrm{i} \sum_j R_j$$

where the R_j are the residues at the poles of $f(z)$ within C.

Solution. Page 859.

Two steps. First is to prove the case that the function f has only a pole of order m at $z = z_0$. Second is extending the above argument to the case of where f is continuous within and on a closed contour C and analytic, except for a finite number of poles within C

$$\oint_C f(z)\mathrm{d}z = 2\pi\mathrm{i} \sum_j R_j$$

Problem 7. Let $b > a > 0$. Evaluate the integral

$$I = \int_0^{2\pi} \frac{\cos 2\theta}{a^2 + b^2 - 2ab\cos\theta} \mathrm{d}\theta.$$

Solution. Page 861,862. Subsection 24.13.1

$$\begin{cases} \cos 2\theta = \dfrac{1}{2}(z^2 + z^{-2}) \\ \cos\theta = \dfrac{1}{2}(z + z^{-1}) \\ \mathrm{d}\theta = -\mathrm{i}z^{-1}\mathrm{d}z \\ C : z = \exp \mathrm{i}\theta \end{cases}$$

Problem 8. Let a be a real number. Evaluate the integral

$$I = \int_0^{\infty} \frac{\mathrm{d}x}{(x^2 + a^2)^4}.$$

Solution. Page 863

Now give a different method to find the residue.

$$\mathrm{Res}\left(\frac{1}{(z^2 + a^2)^4}, |a|\mathrm{i}\right) = \lim_{z \to |a|\mathrm{i}} \frac{1}{3!} \frac{\mathrm{d}^3}{\mathrm{d}z^3} \left[\frac{(z - |a|\mathrm{i})^4}{(z - |a|\mathrm{i})^4 (z + |a|\mathrm{i})^4}\right] = \frac{-5\mathrm{i}}{32|a|^7}$$

Therefore $I = 5\pi/(32|a|^7)$.

Problem 9. Let a be a real number and $m > 0$. Find the principal value of

$$\int_{-\infty}^{\infty} \frac{\cos mx}{x - a} dx.$$

Solution. Omitted

Problem 10. Let $a > 0$. Evaluate the integral

$$I = \int_0^\infty \frac{\mathrm{d}x}{(x+a)^3\sqrt{x}}.$$

Solution. Page 866.

$$\mathrm{Res}(f, -a) = \lim_{z\to -a} \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}z^2}\left[(z+a)^3 \frac{1}{(z+a)^3\sqrt{z}}\right] = \frac{3}{8}(-a)^{-5/2}$$

And

$$I = \frac{3\pi \mathrm{i}}{8(-a)^{5/2}} = \frac{3\pi \mathrm{e}^{\mathrm{i}\pi/2}}{8a^{5/2}\cdot \mathrm{e}^{5\mathrm{i}\pi/2}} = \frac{3\pi}{8a^{5/2}}.$$

Problem 11. Develop the Taylor series expansion of $f(z) = z^2\mathrm{e}^z$ about the point $z_0 = 1$ and determine the value of the fourth derivative $f^{(4)}(1)$.

Solution. Substitute $t = z - 1$, i.e.$z = t + 1$.

$$g(t) = f(t+1) = (t+1)^2\mathrm{e}^{t+1} = \mathrm{e}(t^2+2t+1)\sum_{n=0}^{+\infty}\frac{t^n}{n!}$$

$$\frac{f^{(4)}(1)}{4!} = \frac{\mathrm{e}}{2!} + \frac{2\mathrm{e}}{3!} + \frac{e}{4!} = \frac{21}{24}\mathrm{e}$$

Hence $f^{(4)}(1) = 21\mathrm{e}$.

数学物理方法作业 7(第八周, 练习 2)

唐嘉琪 (2023 级数学与应用数学 1 班)

2025 年 4 月 24 日

摘要

共形映照, 复积分, Cauchy 定理, Cauchy 积分公式, Taylor 级数与 Laurent 级数, 残数定理, 计算定积分.

题目 1. 求一个分式线性变换 $w = \dfrac{az+b}{cz+d}$ 把上半平面映照为自身, 且

$$w(\mathrm{i}) = \mathrm{i},\ \arg w'(\mathrm{i}) = \frac{\pi}{3}.$$

解答. 由 $w(\mathrm{i}) = \mathrm{i}$ 有

$$\frac{a\mathrm{i}+b}{c\mathrm{i}+d} = \mathrm{i} \Longrightarrow a\mathrm{i}+b = -c+d\mathrm{i}.$$

这里断言 $a, b, c, d \in \mathbb{R}$, 由于其将上半平面映为自身, 则一定将实轴映为实轴 (考虑任意实数代入). 比较实部与虚部有 $(a, b) = (d, -c)$. 即有

$$w = \frac{az-c}{cz+a}, \quad (a, c \in \mathbb{R}, a^2+c^2 \neq 0).$$

考察

$$w'(z) = \frac{ad-bc}{(cz+d)^2} = \frac{a^2+c^2}{(cz+a)^2} = \frac{a^2+c^2}{(a^2+c^2)\mathrm{e}^{\mathrm{i}2\theta}} = \mathrm{e}^{-\mathrm{i}2\theta}, \quad \theta = \arctan\left(\frac{c}{a}\right).$$

由条件 $w'(\mathrm{i}) = \pi/3$ 立刻得

$$-2\theta = \frac{\pi}{3} + 2k\pi.$$

取到 $(\theta, a, c) = (-\pi/6, \sqrt{3}, -1)$. 则

$$w = \frac{\sqrt{3}z+1}{-z+\sqrt{3}}.$$

题目 2. 求共形映照 $w = f(z)$, 将第一象限映到单位圆内部, 且

$$f(1+i) = 0, f(\mathrm{i}) = \mathrm{i}.$$

解答. 我们希望求解 $f\colon \{\operatorname{Im} z > 0, \operatorname{Re} z > 0\} \twoheadrightarrow \{|z| < 1\}$. 可以将其拆解为两部分

$$f = g \circ h, \quad h\colon \{\operatorname{Im} z > 0, \operatorname{Re} z > 0\} \twoheadrightarrow \{\operatorname{Im} z > 0\}, g\colon \{\operatorname{Im} z > 0\} \twoheadrightarrow \{|z| < 1\}.$$

h 是容易构造的:$z \overset{h}{\mapsto} z^2$.
g 有标准的形式:$z \overset{g}{\mapsto} \exp \mathrm{i}\phi \dfrac{z - z_0}{z - \bar{z}_0}, \quad z_0 \in \{\operatorname{Im} z > 0\}$.
那么我们可以给出 f 的形式:

$$f(z) = \exp \mathrm{i}\phi \frac{z^2 - z_0}{z^2 - \bar{z}_0}$$

代入条件得到

$$\begin{cases} \exp \mathrm{i}\phi \dfrac{(1+\mathrm{i})^2 - z_0}{(1+\mathrm{i})^2 - \bar{z}_0} = 0 \\ \exp \mathrm{i}\phi \dfrac{\mathrm{i}^2 - z_0}{\mathrm{i}^2 - \bar{z}_0} = \mathrm{i} \end{cases} \Longrightarrow \begin{cases} z_0 = 2\mathrm{i} \\ \exp \mathrm{i}\phi = \dfrac{4}{5} - \dfrac{3}{5}\mathrm{i} \end{cases} \Longrightarrow f(z) = \left(\frac{4}{5} - \frac{3}{5}\mathrm{i}\right) \frac{z^2 - 2\mathrm{i}}{z^2 + 2\mathrm{i}}.$$

题目 3. 计算复积分

$$\oint_C \frac{z + \bar{z}}{z\bar{z}} \mathrm{d}z.$$

其中 C 为正向圆周 $|z| = 3$.

解答. 做替换 $z = 3\mathrm{e}^{\mathrm{i}\theta}$.

$$\begin{aligned} \oint_C \frac{z + \bar{z}}{z\bar{z}} \mathrm{d}z &= \int_0^{2\pi} \frac{3\mathrm{e}^{\mathrm{i}\theta} + 3\mathrm{e}^{-\mathrm{i}\theta}}{3^2 \mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}} \cdot 3\mathrm{i}\mathrm{e}^{\mathrm{i}\theta} \mathrm{d}\theta = \mathrm{i} \int_0^{2\pi} \frac{\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}}{\mathrm{e}^{-\mathrm{i}\theta}} \mathrm{d}\theta \\ &= \mathrm{i} \int_0^{2\pi} (\mathrm{e}^{2\mathrm{i}\theta} + 1) \mathrm{d}\theta = \mathrm{i} \left(\frac{\mathrm{e}^{2\mathrm{i}\theta}}{2\mathrm{i}} + \theta \right) \Bigg|_0^{2\pi} = 2\pi\mathrm{i}. \end{aligned}$$

题目 4. 计算复积分

$$\oint_C (|z| - z\sin(z^2)) \mathrm{d}z.$$

其中 C 是连接 -1 到 1 的直线段, 再从 1 由下半单位圆到 -1.

解答.

$$\oint_C (|z| - z\sin(z^2)) \mathrm{d}z = \oint_C |z| \mathrm{d}z - \oint_C z\sin(z^2) \mathrm{d}z.$$

由 Cauchy 定理立知

$$\oint_C z\sin(z^2)\mathrm{d}z = 0$$

故

$$\oint_C (|z| - z\sin(z^2))\mathrm{d}z = \oint_C |z|\mathrm{d}z = \int_{-1}^{1} |x|\mathrm{d}x + \int_{\text{半圆}} \mathrm{d}z = 1 + \int_{-1}^{1} \mathrm{d}x = -1.$$

题目 5. 计算复积分

$$\oint_C \frac{1}{(z^2+1)(z-3)^2}\mathrm{d}z.$$

其中 C 为正向圆周 $|z| = 2$.

解答. 注意 $f := \dfrac{1}{(z^2+1)(z-3)^2}$ 在 C 内由两个一阶极点 $z = \pm\mathrm{i}$.

$$(z-\mathrm{i})\frac{1}{(z-\mathrm{i})(z+\mathrm{i})(z-3)^2} = \frac{1}{(z+\mathrm{i})(z-3)^2} \to \frac{1}{2\mathrm{i}(\mathrm{i}-3)^2} = \mathrm{Res}(f,\mathrm{i}), \quad z \to \mathrm{i}.$$

$$(z+\mathrm{i})\frac{1}{(z-\mathrm{i})(z+\mathrm{i})(z-3)^2} = \frac{1}{(z-\mathrm{i})(z-3)^2} \to \frac{1}{-2\mathrm{i}(\mathrm{i}+3)^2} = \mathrm{Res}(f,-\mathrm{i}), \quad z \to -\mathrm{i}.$$

那么由残数定理直接有

$$\oint_C \frac{1}{(z^2+1)(z-3)^2}\mathrm{d}z = 2\pi\mathrm{i}\left(\frac{1}{2\mathrm{i}(\mathrm{i}-3)^2} - \frac{1}{2\mathrm{i}(\mathrm{i}+3)^2}\right) = \frac{3}{25}\pi\mathrm{i}.$$

题目 6. 计算复积分

$$\oint_C \frac{z^5-1}{z^5+1}dz.$$

其中 C 为正向圆周 $|z| = 2$.

解答. 显然有一阶极点 $\mathrm{e}^{i\frac{1+2k}{5}\pi} =: z_k, \quad k = 0, 1, \cdots, 4.$

$$\begin{aligned}\oint_C \frac{z^5-1}{z^5+1}dz &= \oint_C \mathrm{d}z - 2\oint_C \frac{1}{z^5+1}dz = -2\oint_C \frac{1}{z^5+1}dz \\ &= -4\pi\mathrm{i}\sum_{k=0}^{4}\mathrm{Res}\left(\frac{1}{z^5+1}, z_k\right).\end{aligned} \tag{1}$$

注意到 $(z^5+1)' = 5z^4$, 且 $5z_0^4 \neq 0$. 那么

$$\mathrm{Res}\left(\frac{1}{z^5+1}, z_k\right) = \frac{1}{5z_k^4}$$

故

$$(1) = \frac{4}{5}\pi\mathrm{i}\sum_{k=0}^{4}\frac{-1}{z_k^4} = \frac{4}{5}\pi\mathrm{i}\sum_{k=0}^{4} z_k = 0.$$

题目 7. 设函数 $f(z)$ 在区域 $|z-z_0|<R$ 内除二阶极点 z_0 外解析且 $f(z)\neq 0$. 设 $0<r<R$, 设 C 为正向圆周 $|z-z_0|=r$. 证明

$$\oint_C \frac{f'(z)}{f(z)}\mathrm{d}z=-4\pi\mathrm{i}.$$

解答. f 在 $|z-z_0|<R$ 内可写为 $f(z)=(z-z_0)^{-2}g(z)$, 满足 $g(z)$ 在 $|z-z_0|<R$ 上非零全纯.

$$\begin{aligned}f'(z)=&-2(z-z_0)^{-3}g(z)+(z-z_0)^{-2}g'(z)\\ \Longrightarrow \oint_C \frac{f'(z)}{f(z)}\mathrm{d}z=&\oint_C\left[\frac{-2}{z-z_0}+\frac{g'(z)}{g(z)}\right]\mathrm{d}z=-2\oint_C\frac{1}{z-z_0}\mathrm{d}z=-4\pi\mathrm{i}.\end{aligned}$$

□

题目 8. 设 $f(z)=\dfrac{1}{z^3-2z^2+z}$, 分别在环域 $0<|z-1|<1$ 和 $|z-1|>1$ 内将 $f(z)$ 展开为 $z-1$ 的洛朗级数, 并计算留数 $\mathrm{Res}(f(z),-1)$.

解答.

$$f(z)=\frac{1}{z(z-1)^2}=\frac{1}{[(z-1)+1](z-1)^2}=\frac{1}{(t+1)t^2},\quad t:=z-1.$$

此时环域变为 $0<|t|<1$ 与 $|t|>1$.

在 $0<|t|<1$ 上,

$$\frac{1}{t+1}=\sum_{n=0}^{\infty}(-1)^n t^n\Rightarrow f(z)=\sum_{n=0}^{\infty}(-1)^n t^{n-2}=\sum_{n\geqslant -2}(-1)^n(z-1)^n.$$

在 $|t|>1$, 即 $0<|1/t|<1$ 上,

$$\frac{1}{t+1}=\frac{1}{t(1+1/t)}=\frac{1}{t}\sum_{n=0}^{\infty}(-1)^n t^{-n}=\sum_{n=0}^{\infty}(-1)^n t^{-(n+1)}.$$

从而

$$f(z)=\sum_{n=0}^{\infty}(-1)^n t^{-n-3}=\sum_{n\geqslant -3}(-1)^{k+1}(z-1)^{-k}.$$

注意到 f 在 -1 处全纯, 故 $\mathrm{Res}(f(z),-1)=0$.

题目 9. 使用留数计算定积分

$$\int_0^\infty \frac{x\sin x}{(x^2+4)^2}\mathrm{d}x.$$

解答.

$$\begin{aligned}\int_0^\infty \frac{x\sin x}{(x^2+4)^2}\mathrm{d}x =&\frac{1}{2}\operatorname{Im}\left[\int_{-\infty}^\infty \mathrm{e}^{\mathrm{i}x}\frac{x}{(x^2+4)^2}\mathrm{d}x\right]\\ =&\frac{1}{2}\operatorname{Im}\left\{2\pi\mathrm{i}\operatorname{Res}\left[\frac{z\mathrm{e}^{\mathrm{i}z}}{(z^2+4)^2},2\mathrm{i}\right]\right\}.\end{aligned} \tag{2}$$

注意二阶奇点 2i.

$$\begin{aligned}\operatorname{Res}\left[\frac{z\mathrm{e}^{\mathrm{i}z}}{(z^2+4)^2},2\mathrm{i}\right] &= \lim_{z\to 2\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}z}\frac{z\mathrm{e}^{\mathrm{i}z}}{(z+2\mathrm{i})^2}\\ &= \lim_{z\to 2\mathrm{i}}\frac{(\mathrm{e}^{\mathrm{i}z}+\mathrm{i}z\mathrm{e}^{\mathrm{i}z})(z+2\mathrm{i})^2 - z\mathrm{e}^{\mathrm{i}z}\cdot 2(z+2\mathrm{i})}{(z+2\mathrm{i})^4} = \frac{1}{8\mathrm{e}^2}\end{aligned}$$

$$\Longrightarrow (2) = \frac{1}{2}\operatorname{Im}\left(2\pi\mathrm{i}\cdot\frac{1}{8\mathrm{e}^2}\right) = \frac{\pi}{8\mathrm{e}^2}.$$

题目 10. 将函数 $f(z)=\dfrac{1}{z^7(1-z^2)^2}$ 在圆环区域 $1<|z|<\infty$ 展开成洛朗级数.

解答.

$$\begin{aligned}\frac{1}{(1-z^2)^2} =&\frac{1}{z^4}\cdot\frac{1}{(1-1/z^2)^2} = \frac{1}{z^4}\sum_{n=1}^\infty n\left(\frac{1}{z^2}\right)^{n-1}, \quad 0<|1/z|<1.\\ \Longrightarrow f(z) =&z^{-7}\cdot z^{-4}\cdot\sum_{n=1}^\infty nz^{2-2n} = \sum_{n\geqslant 0}(n+1)z^{-2n-7}\end{aligned}$$

题目 10 的注记.

$$\frac{1}{1-w} = \sum_{n=1}^\infty nw^{n-1}, \quad 0<|w|<1.$$

题目 11. 将函数 $f(z)=\dfrac{1}{z^4-5z^2+4}$ 在圆环区域 $1<|z|<2$ 展开成洛朗级数.

解答.

$$f(z) = \frac{1}{(z+1)(z-1)(z+2)(z-2)} = -\frac{1}{6}\frac{1}{z-1}+\frac{1}{6}\frac{1}{z+1}+\frac{1}{12}\frac{1}{z-2}-\frac{1}{12}\frac{1}{z+2}.$$

其中系数是由留数法所确定:

$$\frac{1}{(1-2)(1+2)(1+1)}=-\frac{1}{6},\quad \frac{1}{(-1-2)(-1+2)(-1-1)}=\frac{1}{6},$$
$$\frac{1}{(2-1)(2+1)(2+2)}=\frac{1}{12},\quad \frac{1}{(-2-1)(-2+1)(-2-2)}=-\frac{1}{12}.$$

分别展开

$$\frac{1}{z-1}=\frac{1}{z}\frac{1}{1-1/z}=\sum_{n\geqslant 0}z^{-n-1},\quad \frac{1}{z+1}=\frac{1}{z}\frac{1}{1+1/z}=\sum_{n\geqslant 0}(-1)^n z^{-n-1},\quad (|z|>1).$$
$$\frac{1}{z-2}=-\frac{1}{2}\frac{1}{1-z/2}=-\sum_{n\geqslant 0}\frac{z^n}{2^{n+1}},\quad \frac{1}{z+2}=\frac{1}{2}\frac{1}{1+z/2}=\sum_{n\geqslant 0}(-1)^n\frac{z^n}{2^{n+1}},\quad (|z|<2).$$

于是得

$$\begin{aligned}f(z)=&-\frac{1}{6}\sum_{n\geqslant 0}\frac{1}{z^{n+1}}+\frac{1}{6}\sum_{n\geqslant 0}\frac{(-1)^n}{z^{n+1}}-\frac{1}{12}\sum_{n\geqslant 0}\frac{z^n}{2^{n+1}}-\frac{1}{12}\sum_{n\geqslant 0}(-1)^n\frac{z^n}{2^{n+1}}\\=&\frac{1}{6}\sum_{n\geqslant 0}\frac{-1+(-1)^n}{z^{n+1}}-\frac{1}{12}\sum_{n\geqslant 0}\frac{z^n+(-1)^n z^n}{2^{n+1}},\quad (1<|z|<2).\end{aligned}$$

题目 12. 使用留数计算定积分

$$\int_0^{\pi/2}\frac{1}{1+\cos^2 x}\mathrm{d}x.$$

解答.

$$\begin{aligned}\int_0^{\pi/2}\frac{1}{1+\cos^2 x}\mathrm{d}x=&\frac{1}{4}\int_0^{2\pi}\frac{1}{1+\cos^2 x}\mathrm{d}x=\frac{1}{4}\oint_{|z|=1}\frac{1}{1+\left[\frac{1}{2}(z+1/z)\right]^2}\cdot(-\mathrm{i}z^{-1})\mathrm{d}z\\=&-\mathrm{i}\oint_{|z|=1}\frac{z}{z^4+6z^2+1}\mathrm{d}z,\quad f:=\frac{z}{z^4+6z^2+1}\\=&2\pi\left(\mathrm{Res}\left(f,\mathrm{i}(\sqrt{2}-1)\right)+\mathrm{Res}\left(f,-\mathrm{i}(\sqrt{2}-1)\right)\right)\\=&2\pi\left(\frac{1}{8\sqrt{2}}+\frac{1}{8\sqrt{2}}\right)=\frac{\pi}{2\sqrt{2}}.\end{aligned}$$

题目 13. 设函数 $f(z)=\dfrac{e^z}{\cos z}$ 的泰勒展开式为 $\sum\limits_{n=1}^{\infty}c_n z^n$, 求这个幂级数的收敛半径.

解答.

$$\cos z=0\Longrightarrow z=\frac{\pi}{2}+k\pi,\quad k\in\mathbb{Z}.$$

从而收敛半径 $R=\pi/2$.

题目 14. 设幂级数 $\sum\limits_{n=1}^{\infty} c_n(z-2)^n$ 在 $z=4$ 收敛而在 $z=2+2i$ 发散, 求这个幂级数的收敛半径.

解答. 注意到在 $|z-2|=2$ 上 $z=4$ 处收敛, $z=2+2\mathrm{i}$ 处发散. 而只有在 $|z-2|=R$ 上有这种性质 (既有收敛点又有发散点). 故收敛半径 $R=2$.

题目 15. 设 C 是单位圆周 $|z|=1$, 求积分

$$\oint_C \frac{e^z}{z}\mathrm{d}z.$$

由此证明

$$\int_0^{\pi} \mathrm{e}^{\cos t}\cos(\sin(t))\mathrm{d}t=\pi.$$

解答.

$$\oint_{|z|=1} \frac{\mathrm{e}^z}{z}\mathrm{d}z = 2\pi\mathrm{i}\,\mathrm{Res}\left(\frac{\mathrm{e}^z}{z},0\right)=2\pi\mathrm{i}\lim_{z\to 0}\mathrm{e}^z=2\pi\mathrm{i}.$$

现在来证明 $\int_0^{\pi} \mathrm{e}^{\cos t}\cos(\sin(t))\mathrm{d}t=\pi$.

$$\oint_C \frac{\mathrm{e}^z}{z}\mathrm{d}z=\int_0^{2\pi}\frac{\mathrm{e}^{\mathrm{e}^{\mathrm{i}\theta}}}{\mathrm{e}^{\mathrm{i}\theta}}\mathrm{i}\mathrm{e}^{\mathrm{i}\theta}\mathrm{d}\theta=\mathrm{i}\int_0^{2\pi}\mathrm{e}^{\mathrm{e}^{\mathrm{i}\theta}}\mathrm{d}\theta\Rightarrow\int_0^{2\pi}\mathrm{e}^{\mathrm{e}^{\mathrm{i}\theta}}\mathrm{d}\theta=2\pi.$$

展开 $\mathrm{e}^{\mathrm{e}^{\mathrm{i}\theta}}$.

$$\mathrm{e}^{\mathrm{e}^{\mathrm{i}\theta}}=\mathrm{e}^{\cos\theta+\mathrm{i}\sin\theta}=\mathrm{e}^{\cos\theta}\left(\cos(\sin\theta)+\mathrm{i}\sin(\sin\theta)\right)\Rightarrow\int_0^{2\pi}\mathrm{e}^{\cos\theta}\cos(\sin\theta)\mathrm{d}\theta=2\pi.$$

下式是显然的

$$\int_0^{2\pi}\mathrm{e}^{\cos\theta}\cos(\sin\theta)\mathrm{d}\theta=2\int_0^{\pi}\mathrm{e}^{\cos\theta}\cos(\sin\theta)\mathrm{d}\theta$$

从而证毕. □

题目 16. 求解方程

$$\frac{1}{2\pi\mathrm{i}}\oint_C\frac{\xi^4+\xi-1}{\xi-z}d\xi=z-2+\mathrm{i}.$$

其中 C 为正向圆周 $|z|=2$.

解答. $\xi^4+\xi-1$ 在 C 上及内部全纯. 若 z 在 C 内, 那么

$$\frac{1}{2\pi \mathrm{i}}\oint_C \frac{\xi^4+\xi-1}{\xi-z}d\xi = z^4+z-1 = z-2+\mathrm{i} \Rightarrow z^4=-1+\mathrm{i}.$$

从而解得

$$z_k=(\sqrt{2})^{1/4}\cdot \mathrm{e}^{\mathrm{i}\pi(\frac{3}{16}+\frac{k}{\pi})},\quad k=0,1,2,3.$$

若 z 在 C 外, 那么

$$\frac{1}{2\pi \mathrm{i}}\oint_C \frac{\xi^4+\xi-1}{\xi-z}d\xi = 0 \Rightarrow z=2-\mathrm{i}.$$

题目 17. 设 $f(z)$ 是整函数, 设存在正整数 n 与常数 M 使得对充分大的 $|z|$, 有 $|f(z)|\leqslant M|z|^n$. 证明 $f(z)$ 是一个次数小于等于 n 的多项式.

解答. 由于 f 是整函数 (在全平面上全纯), 那么 f 在全平面上展开为 Taylor 级数:

$$\sum_{k\geqslant 0} a_k z^k.$$

现在对系数进行估计

$$\begin{aligned}|a_k| = \frac{|f^(k)(0)|}{k!} &\leqslant \frac{1}{k!}\cdot\frac{k!\sup\limits_{|z|=R}|f(z)|}{R^k} = \frac{\sup\limits_{|z|=R}|f(z)|}{R^k}\\ &\leqslant \frac{MR^n}{R^k} = MR^{n-k},\quad \forall k\geqslant 0.\end{aligned}$$

因此对于 $k>n$ 当 $R\to\infty$ 有 $|a_k|\leqslant MR^{n-k}\to 0$. 这表明 Taylor 级数退化为有限和

$$f=\sum_{k=0}^{n} a_k z^k.$$

即 f 是次数不超过 n 的多项式. □

题目 18. 设函数 $f(z)$ 在包含圆盘 $C:|z|\leq R$ 的一个区域内解析, 在 C 上不为零. 证明 $f(z)$ 在 C 内部的零点个数（重数计算在内）等于

$$\frac{1}{2\pi \mathrm{i}}\oint_C \frac{f'(z)}{f(z)}\mathrm{d}z.$$

解答. 设 $z=a$ 是 m 重零点, 则

$$f(z)=(z-a)^m g(z), \quad g(a)\neq 0.$$

现在计算 a 处留数

$$\frac{f'(z)}{f(z)}=\frac{m(z-a)^{m-1}g(z)+(z-a)^m g'(z)}{(z-a)^m g(z)}=\frac{m}{z-a}+\frac{g'(z)}{g(z)}$$

由于 $\dfrac{g'(z)}{g(z)}$ 解析, 那么

$$\operatorname{Res}\left(\frac{f'(z)}{f(z)}, a\right)=m.$$

设 f 在 C 内有 w_i 重零点 z_i, $i\in I$, 那么直接有

$$\operatorname{Res}(\frac{f'(z)}{f(z)}, z_i)=w_i \Rightarrow \frac{1}{2\pi \mathrm{i}}\oint_C \frac{f'(z)}{f(z)}\mathrm{d}z=\sum_{i\in I} w_i.$$

□

题目 19. 设函数 $f(z)$ 在包含圆盘 $D:|z-z_0|\leqslant R$ 的一个区域内解析, 设 $|f(z_0)|=\max\{|f(z)|: z\in D\}$. 证明 $f(z)$ 是常数函数.

解答. f 是域 D 中的非常数的全纯函数, 那么 $|f(z)|$ 不可能在 D 中取到最大值. 从而 f 是常数. 即由最大模原理直接得证 □.

题目 20. 设 $f(z)$ 在复平面内除一点 z_0 外解析, 且 $|f(z)|$ 在 z_0 的领域内有界. 证明 $\int_C f(z)\mathrm{d}z=0$ 对复平面上的任意闭曲线都成立.

解答. 由 $|f(z_0)|$ 在 z_0 的领域内有界, 且 z_0 为孤立奇点, 那么 z_0 只能为可去奇点. 补充定义 z_0 处的值 $f(z_0):=\lim\limits_{z\to z_0} f(z)$. 那么补充后的函数 $\hat{f}$ 在全平面上解析, 自然有 $\int_C \hat{f}\mathrm{d}z=0$. 由于 f 与 $\hat{f}$ 在全平面上只有一个点 z_0 处的值不同, 那么 $\int_C f\mathrm{d}z=\int_C \hat{f}\mathrm{d}z=0$. □